Light as Particles (Photons)

The following subjects will be covered

- The Birth of Quantum Mechanics
- The Photoelectric Effect
- Light Wave-Particle Duality.
- Light as Photons.
- Examples

The Birth of Quantum Mechanics

The process of generating light from heat was very well "known" BUT
the actual physics was not "understood"
The relation between Temperature and Light intensity and wavelength was not developed.



Max Planck
1858-1947

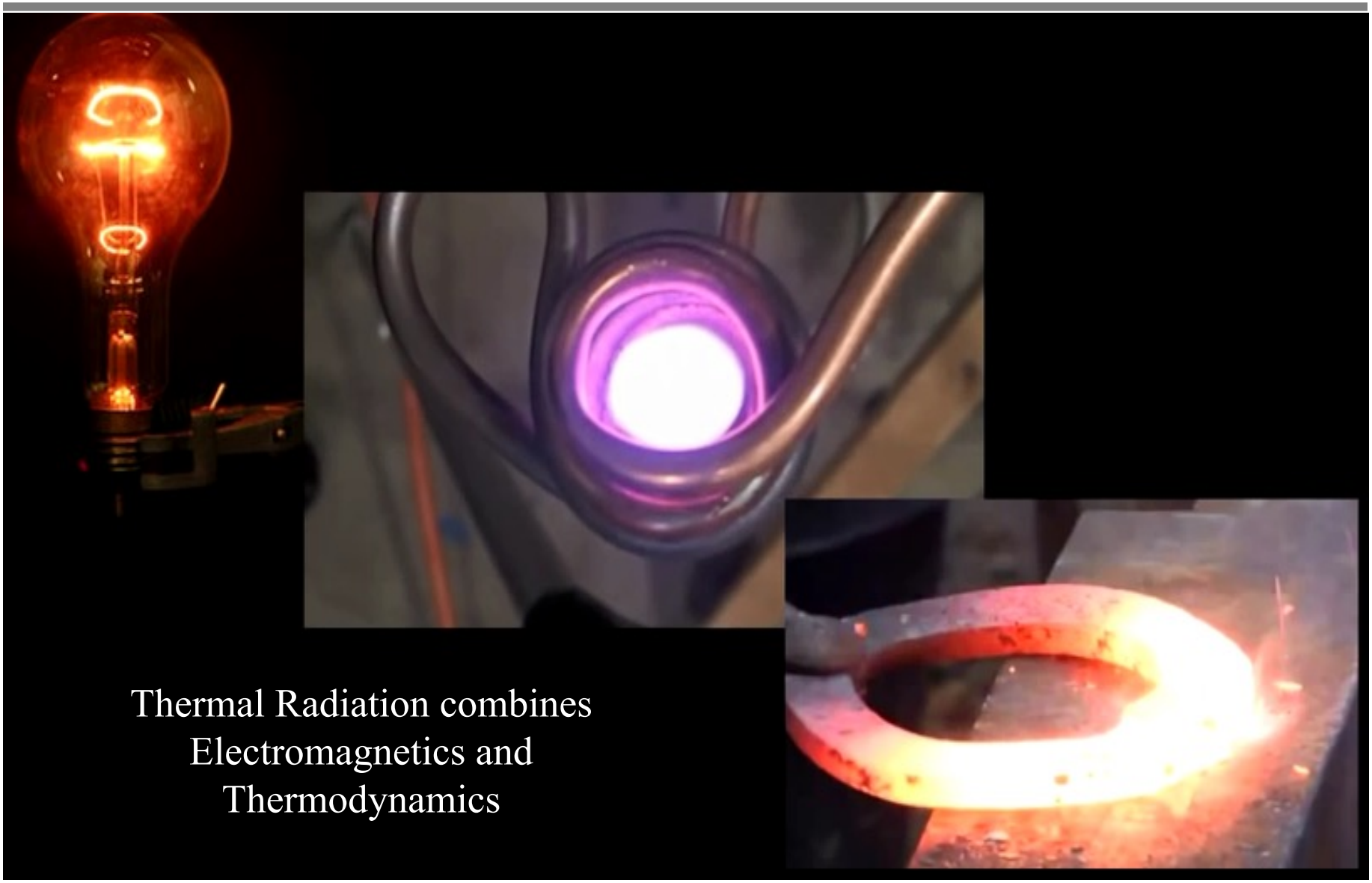
Planck began studying physics in 1874 after being advised against this by a physics professor on the grounds that "*in this field, almost everything is already discovered, and all that remains is to fill a few holes.*"

His doctoral thesis concerned thermodynamics.

In 1894 Planck was commissioned to research how to obtain the most output from a light bulb for a given input of electrical power.

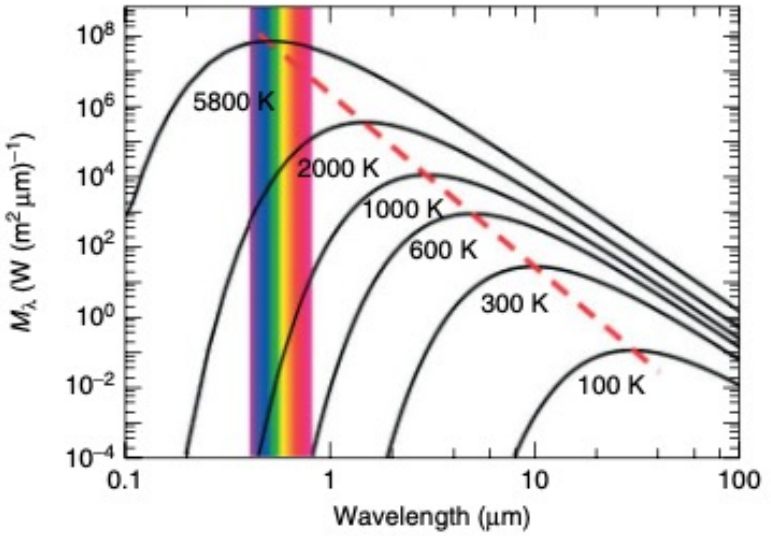


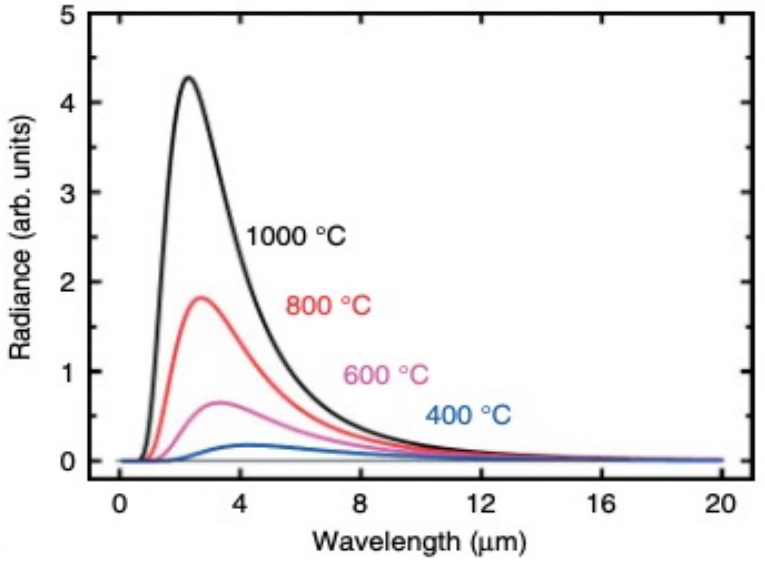
The Birth of Quantum Mechanics

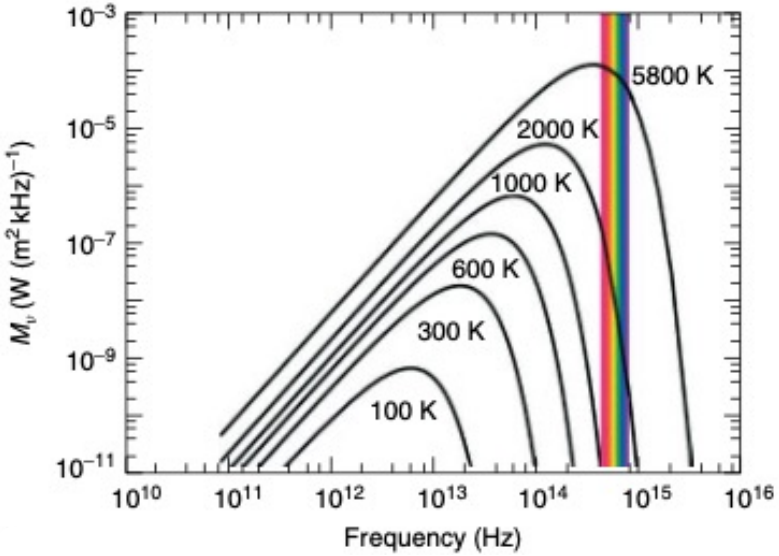


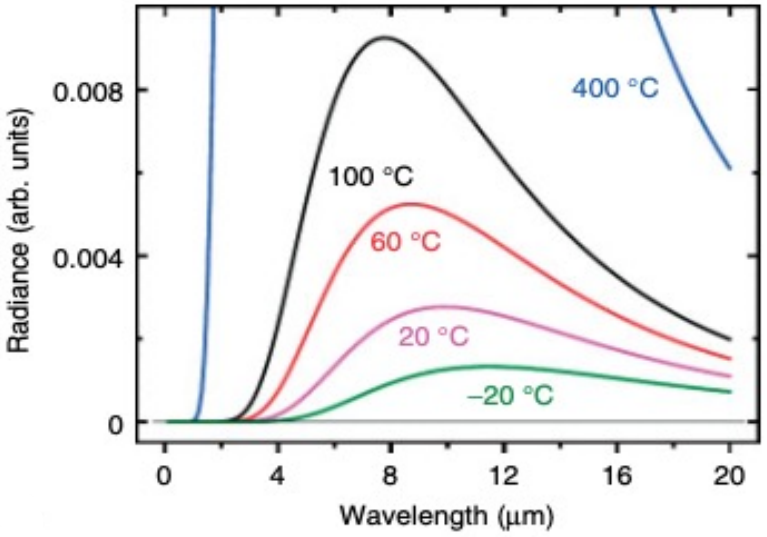
Thermal Radiation combines
Electromagnetics and
Thermodynamics

Thermal Radiation

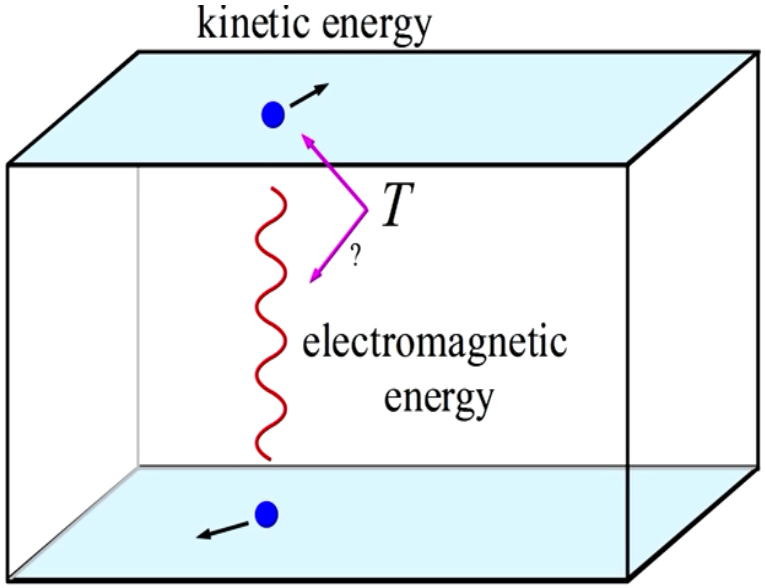








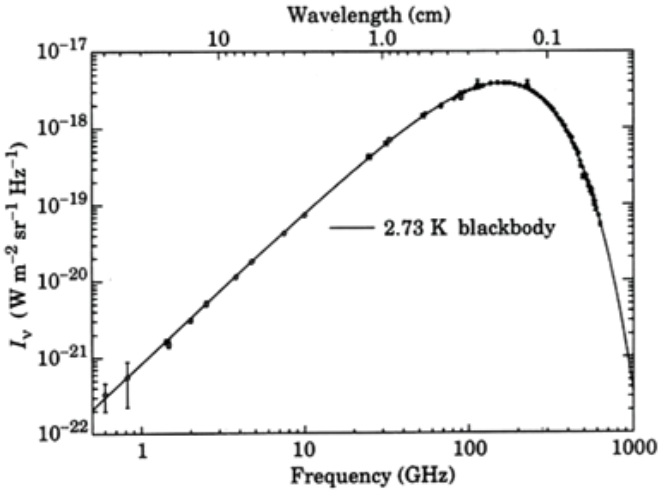
The Birth of Quantum Mechanics



Rayleigh-Jeans law

$$u(\nu) = A\,\nu^2 kT$$

(straight-line on a log-log plot)



Each mode sho Classical physics $u(\nu) = A\,\nu^2 kT$ nergy kT

In thermodynan more modes at $\langle E \rangle_{\text{mode}} = kT$ diation to
increase with te high frequencies

The Birth of Quantum Mechanics

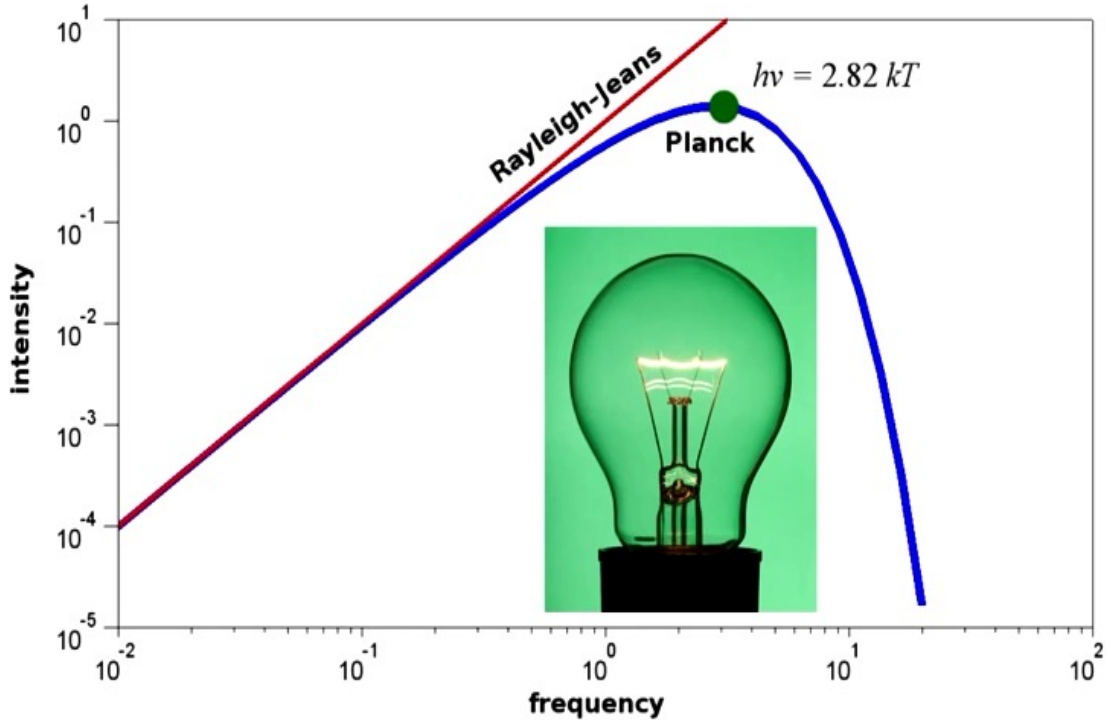
for finite h

$$u(\nu) = A\,\nu^2 \frac{h\nu}{e^{h\nu/kT} - 1}$$

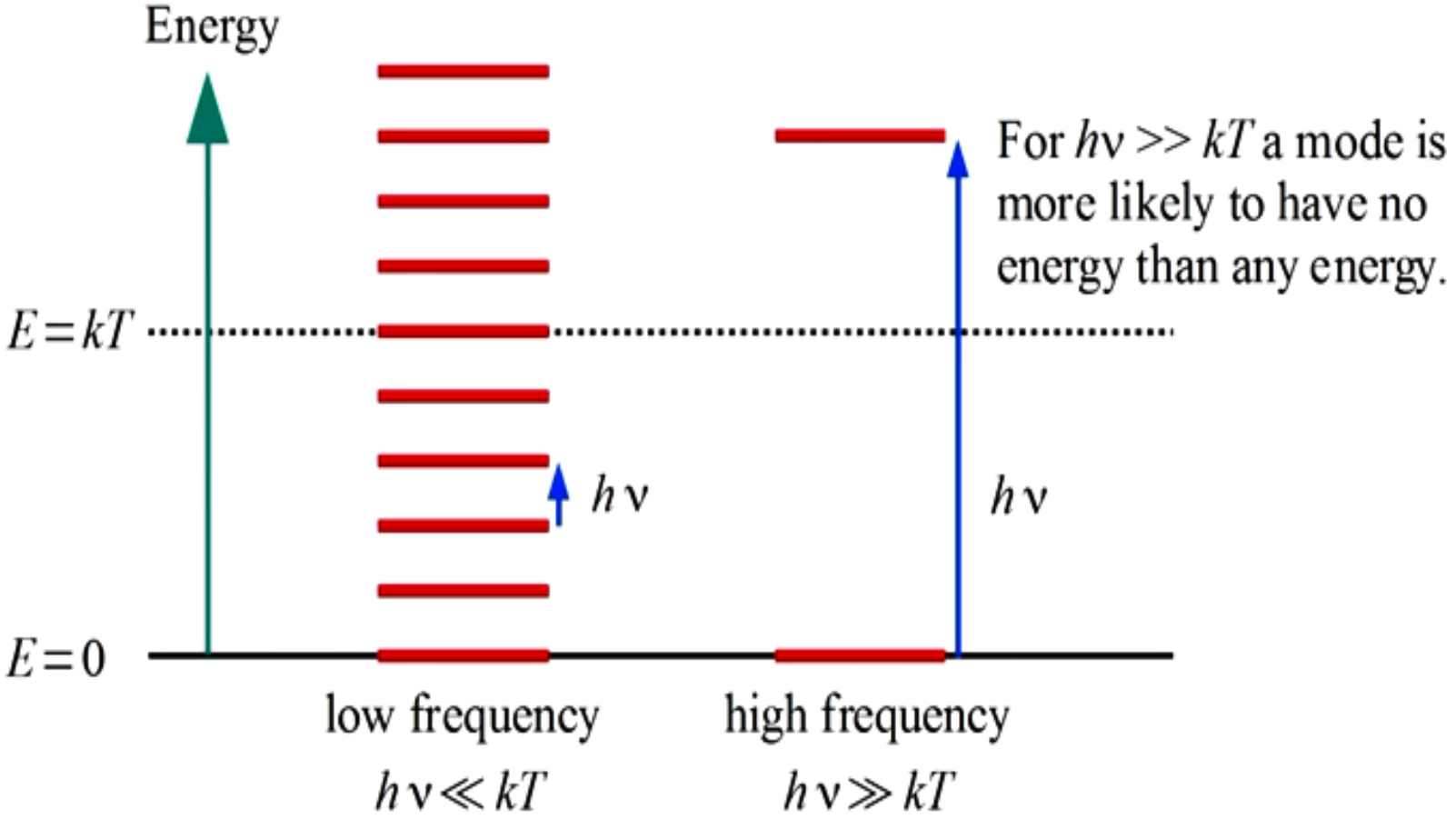
Planck's law (1900)

$$E = n h\nu$$

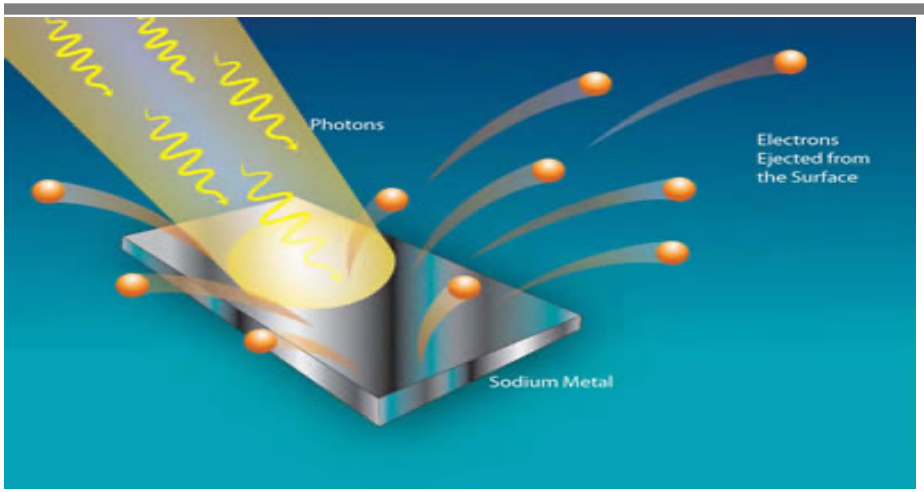
$$h = 6.626 \cdot 10^{-34}\,\text{Js}$$

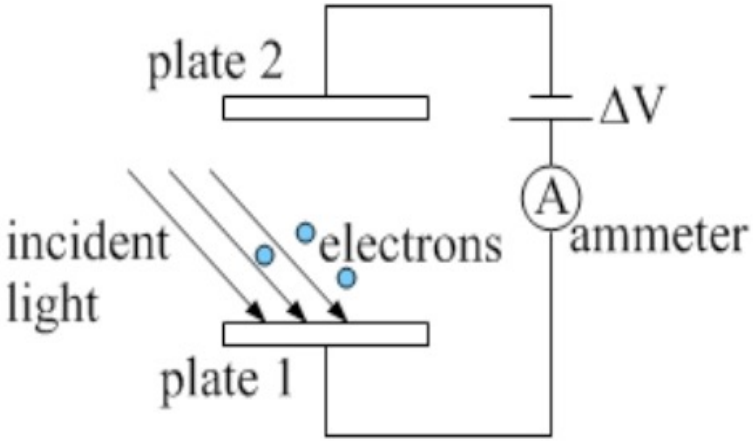


The Birth of Quantum Mechanics



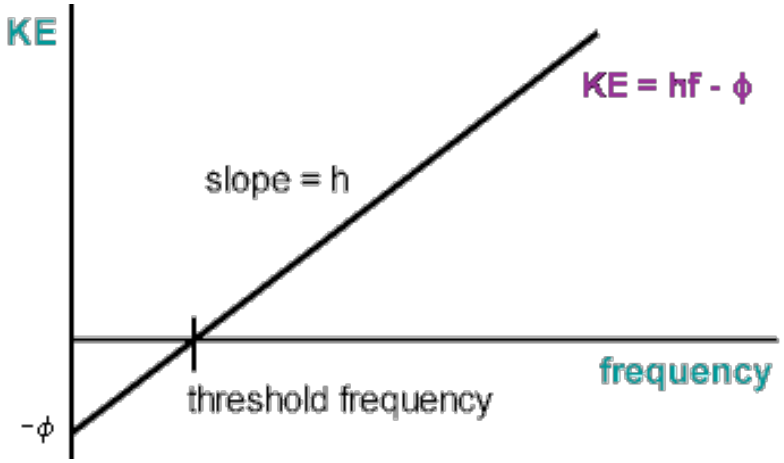
The Photo-electric Effect





Observations

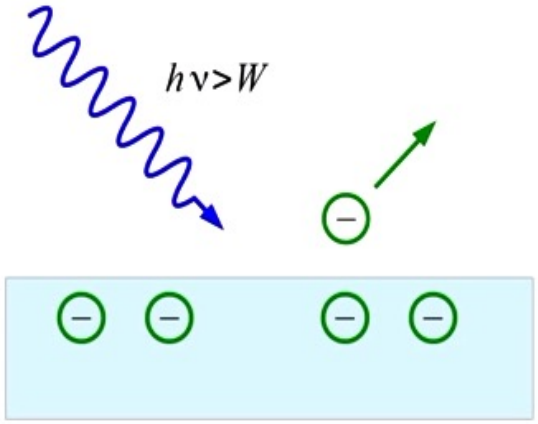
- No current detected for $\omega < \omega_{th}$ even with high intensity.
- Current detected when $\omega > \omega_{th}$ and current increases with light intensity.
- Electron kinetic energy is linear with ω.

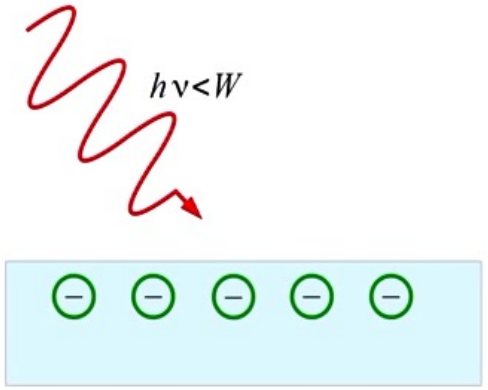


The Photo-electric Effect

Einstein 1905: If W is the energy needed to remove an electron, only photons with $h\nu > W$ will eject electrons. The number of ejected electrons is proportional to the number of photons. The kinetic energy is $h\nu - W$.

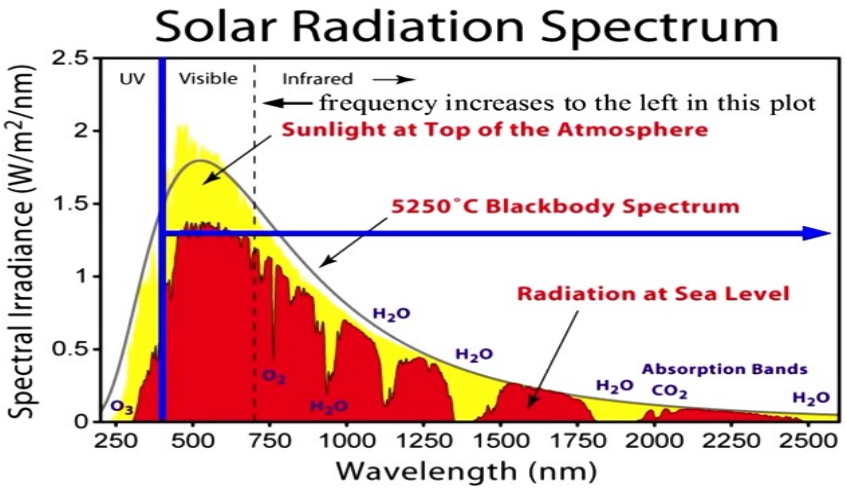


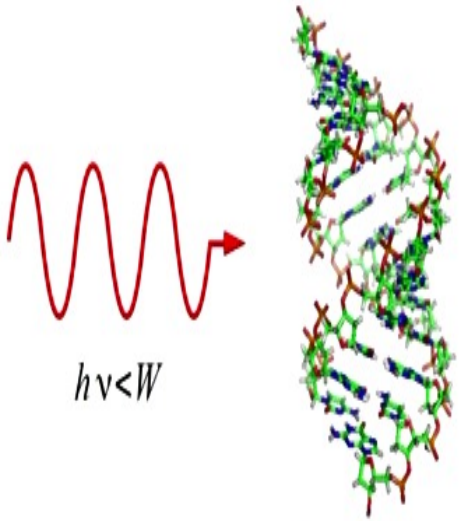




The Birth of Quantum Mechanics



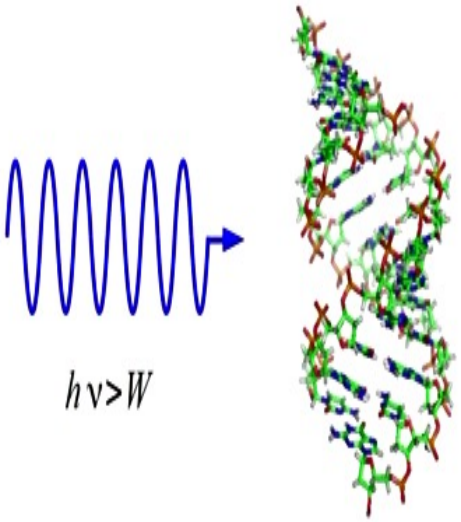




$h\nu < W$

no chemical reaction

damaging reaction
requires energy W



$h\nu > W$

chemical reaction

Wave-particle Duality



The Light Quanta: The Photon

Based on these observations, the photon concept emerges

What is a photon?

A light wave with frequency ω can be thought of as composed of a stream of photons (energy particles) that have the same wave speed (c) and travel in the same wave direction.

Photon Energy $E = hf = \hbar\omega = hc/\lambda$

Photon Momentum $p = \frac{E}{v} = \frac{\hbar\omega}{v} = \frac{h}{\lambda}$

Photon mass = 0

Photon spin = 1

Photon Stream

Average Photon Flux density in a light wave $\varphi = \frac{I}{\hbar\omega}$ $\frac{photons}{s \cdot m^2}$

Average Photon Flux $\int \varphi \, dA$ $\frac{photons}{s}$

Star light 10^6

Moon light 10^8

Sun light 10^{14}

Laser light 10^{22}

$\frac{photons}{s \cdot cm^2}$

Plank's constant
$h = 6.626\times10^{-34}\ Js$

Reduced Plank's constant ($h/2\pi$)
$\hbar = 1.055\times10^{-34}\ Js$

The Light Quanta: The Photon

Example. What is the energy in a radiation mode at $\lambda = 1\ \mu m$, in free space?

$\lambda = 1\ \mu m$ → $\omega = 2\pi\ c/\lambda$

Infrared

$\omega = 2\pi \times 3 \times 10^{14}$ rad/s

Photon Energy $E = \hbar\ \omega = 1.99 \times 10^{-19}\ J$

$$E = \frac{1.99 \times 10^{-19}}{q} = 1.24\ eV$$

Example. What is the photon flux in a 0.2-μm light, if the total power over some area is 1 nW?

$\lambda = 0.2\ \mu m$ → $\hbar\ \omega = 10 \times 10^{-19}\ J$

UV-C

$$\text{Photon flux} = \frac{1 \times 10^{-9}\ J/s}{10 \times 10^{-19}\ J} = 1 \times 10^{9} \quad \frac{photons}{s}$$

Example

Ultraviolet light with a wavelength of 240 nm illuminates plate 1 in the figure. Consequently, electrons are emitted and driven across the gap to plate 2 and hence current is read on the ammeter. The battery voltage is gradually increased until the current drops to zero. The battery voltage at this point is 1.2 V.

Find,

1) The energy of the illuminating light.

$$E = 1.24/0.24 = 5.17\ eV$$

2) Maximum kinetic energy of the electron beam.

$$KE = 1.2\ eV$$

3) The work function of the metal.

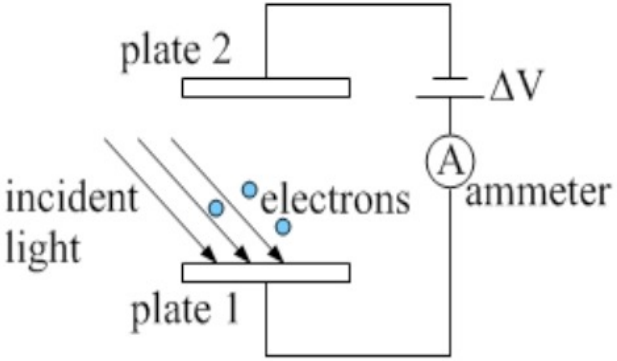
$$W = 5.17 - 1.2 = 3.93\ eV$$

4) The metal used for plate 1.

$$Al$$

5) The maximum wavelength that causes current.

$$\lambda_{max} = ch/\omega = 330\ nm$$



Metal	Work Function, ϕ, (in eV)
Na	2.28
Al	4.08
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14
Fe	4.50

Semiconductors Fundamentals

The following subjects will be covered

- Semiconductors in General.
- The PN Junction.
- The Energy Band Diagram.
- Carrier Statistics in Semiconductors.
- PN Junction Band Diagram.
- Photon Generation and Absorption.
- Direct and Indirect Gap Materials.
- Examples.

Semiconductors

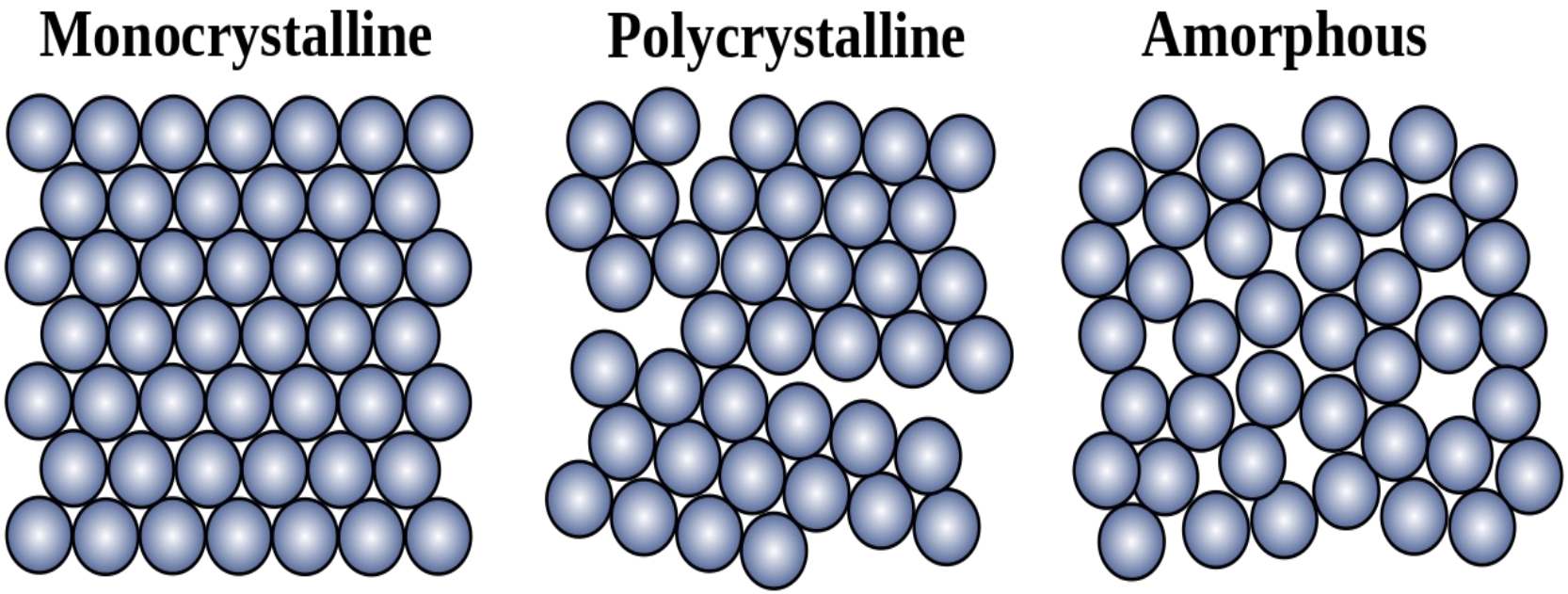
Elemental: Si, Ge.

Compounds: GaAs, InP.

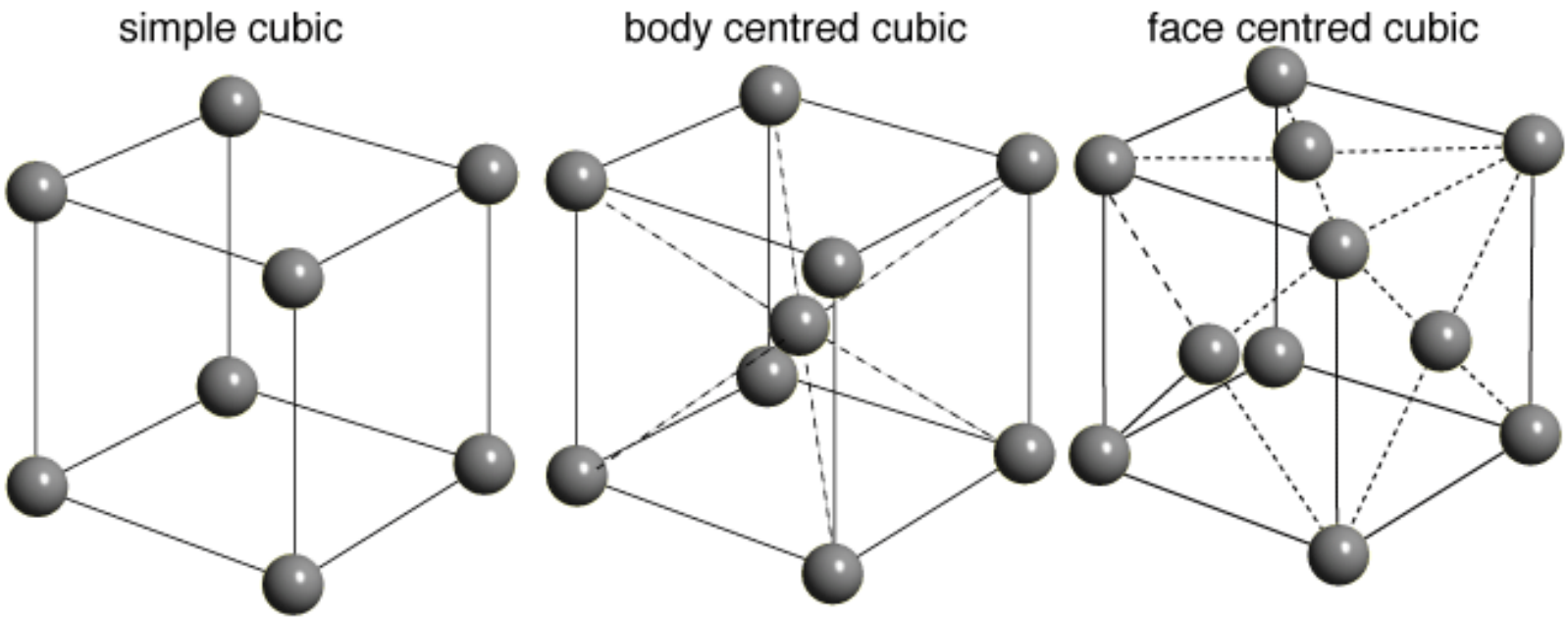
Alloys: $Al_{1-x}Ga_xAs$.

							VIIIA
		IIIA	IVA	VA	VIA	VIIA	2 He 4.003
		5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.183
IB	IIB	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.064	17 Cl 35.453	18 Ar 39.948
29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.922	34 Se 78.96	35 Br 79.909	36 Kr 83.80
47 Ag 107.870	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.904	54 Xe 131.30
79 Au 196.967	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.980	84 Po (210)	85 At (210)	86 Rn (222)

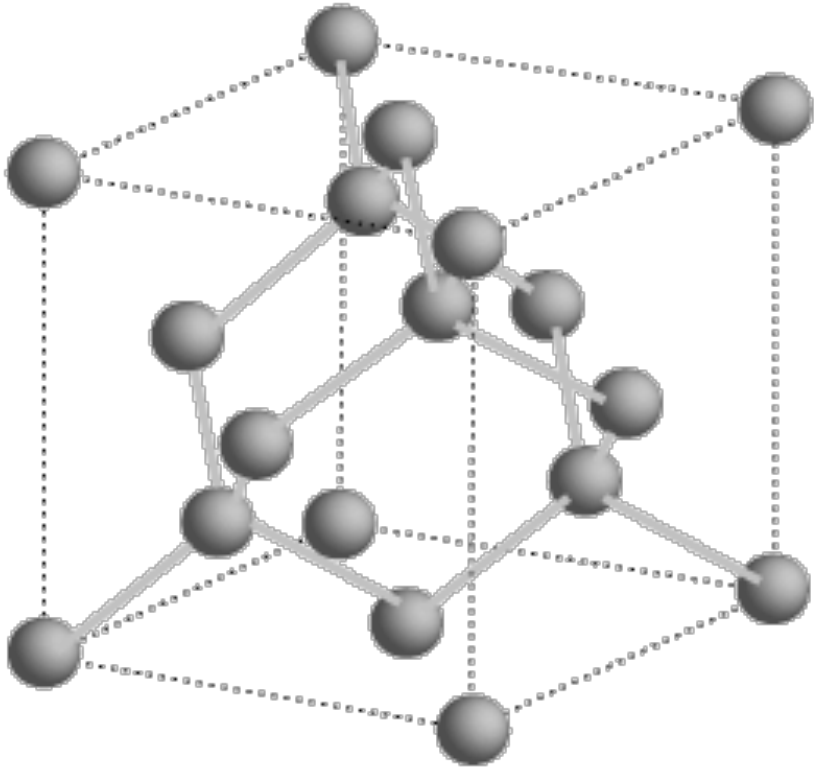
Structure



Crystal Structure: Cubic

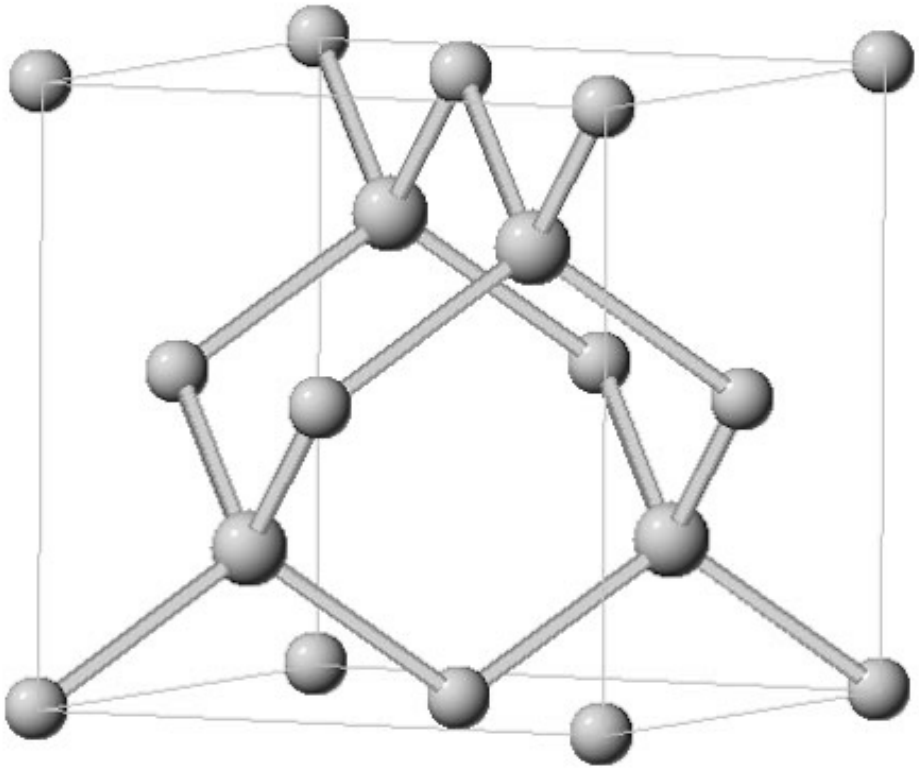


Crystal Structure: Diamond





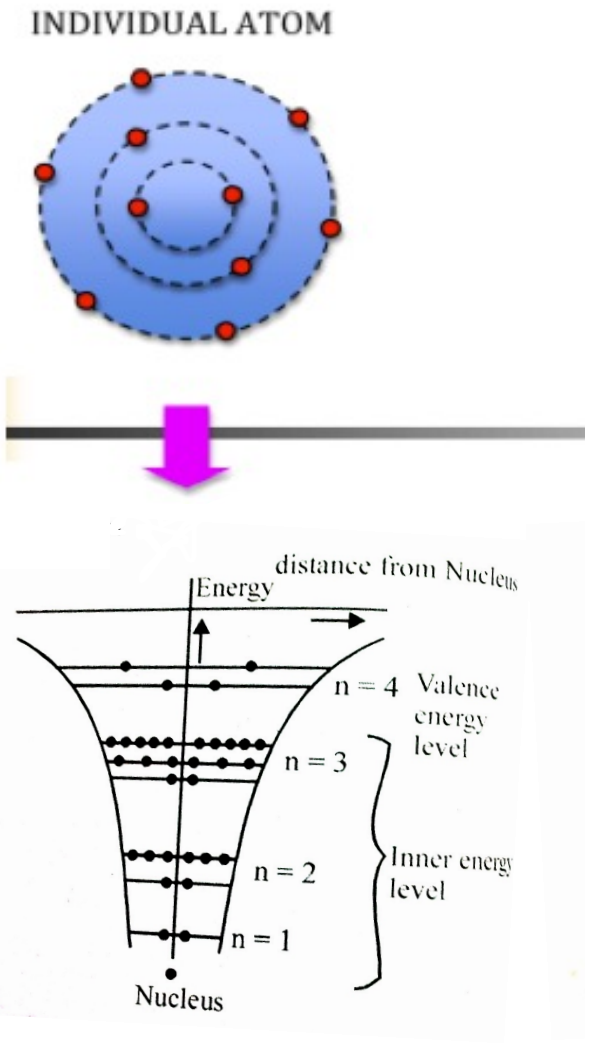
Crystal Structure: Zinc-blende



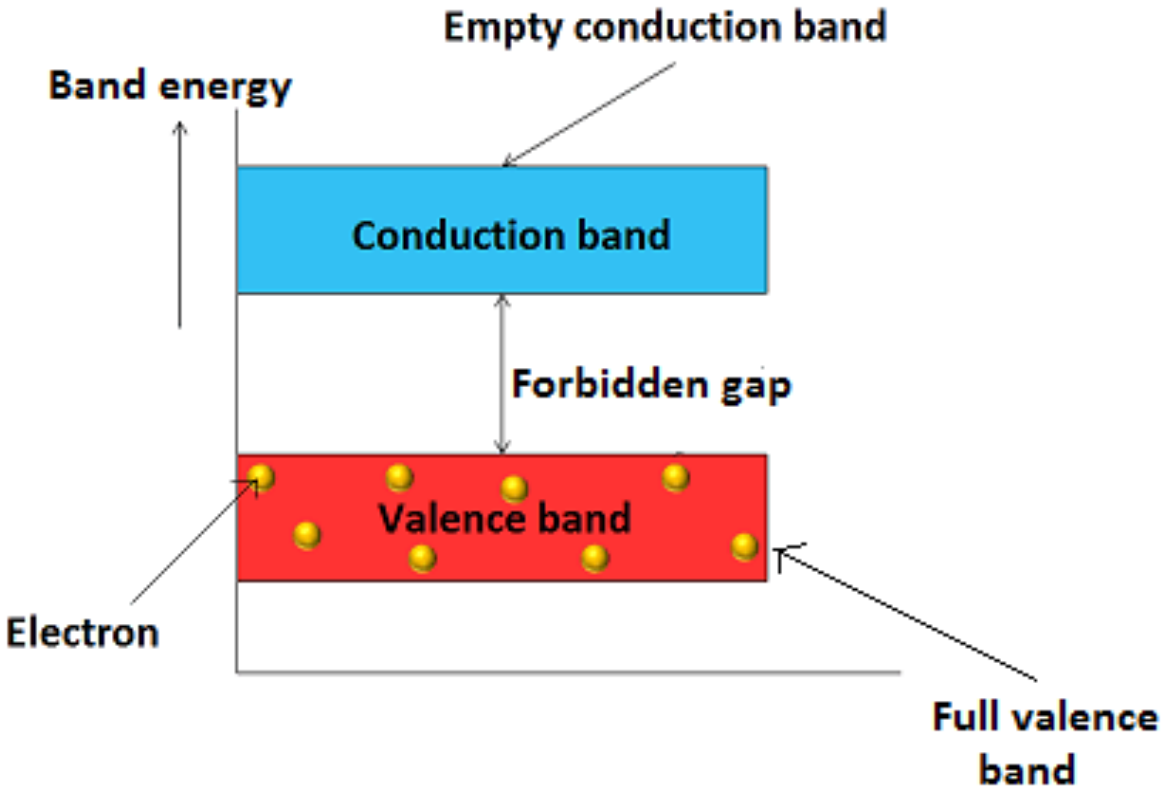
Crystal Structure: Examples

	Crystal	a[Å]	Crystal	a[Å]
Diamond	C	3.567	CuBr	5.691
	Si	5.431	CuCl	5.406
	Ge	5.657	CuI	6.054
	α-Sn	6.491	GaAs	5.654
			GaP	5.451
Zinc-blende	AgI	6.473	GaSb	6.118
	AlAs	5.62	HgS	5.852
	AlP	5.451	HgSe	6.084
	AlSb	6.135	HgTe	6.429
	BAs	4.777	InAs	6.036
	BN	3.615	InP	5.869
	BP	4.538	InSb	6.478
	BeS	4.85	SiC	4.348
	BeSe	5.07	ZnO	4.63
	BeTe	5.54	ZnS	5.409
	CdS	5.818	ZnSe	5.668
	CdSe	6.05	ZnTe	6.089
	CdTe	6.480		

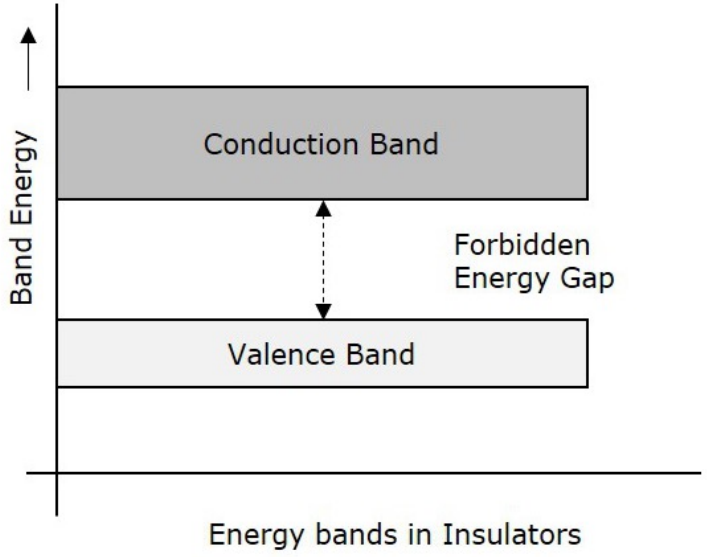
Energy Bands: Isolated atoms and Crystals



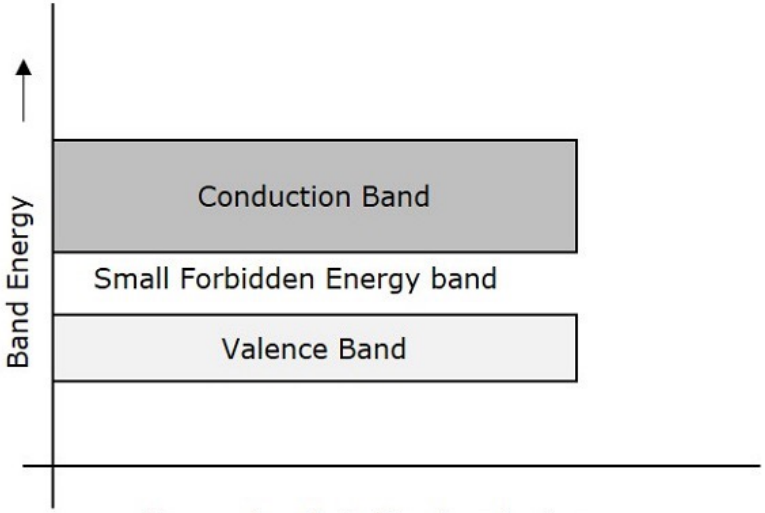
Energy Bands: Isolated atoms and Crystals



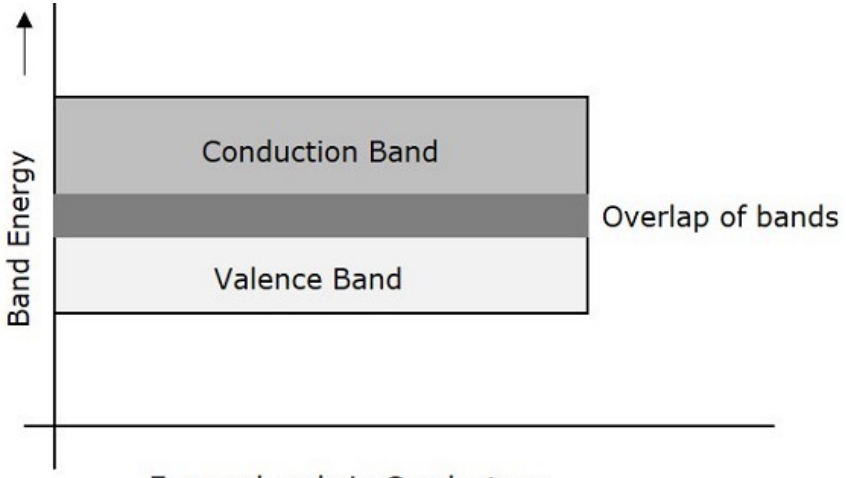
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

Quartz	E_g= 10 eV
Silicon	E_g= 1.12 eV
Copper	E_g= 0 eV

Semiconductors

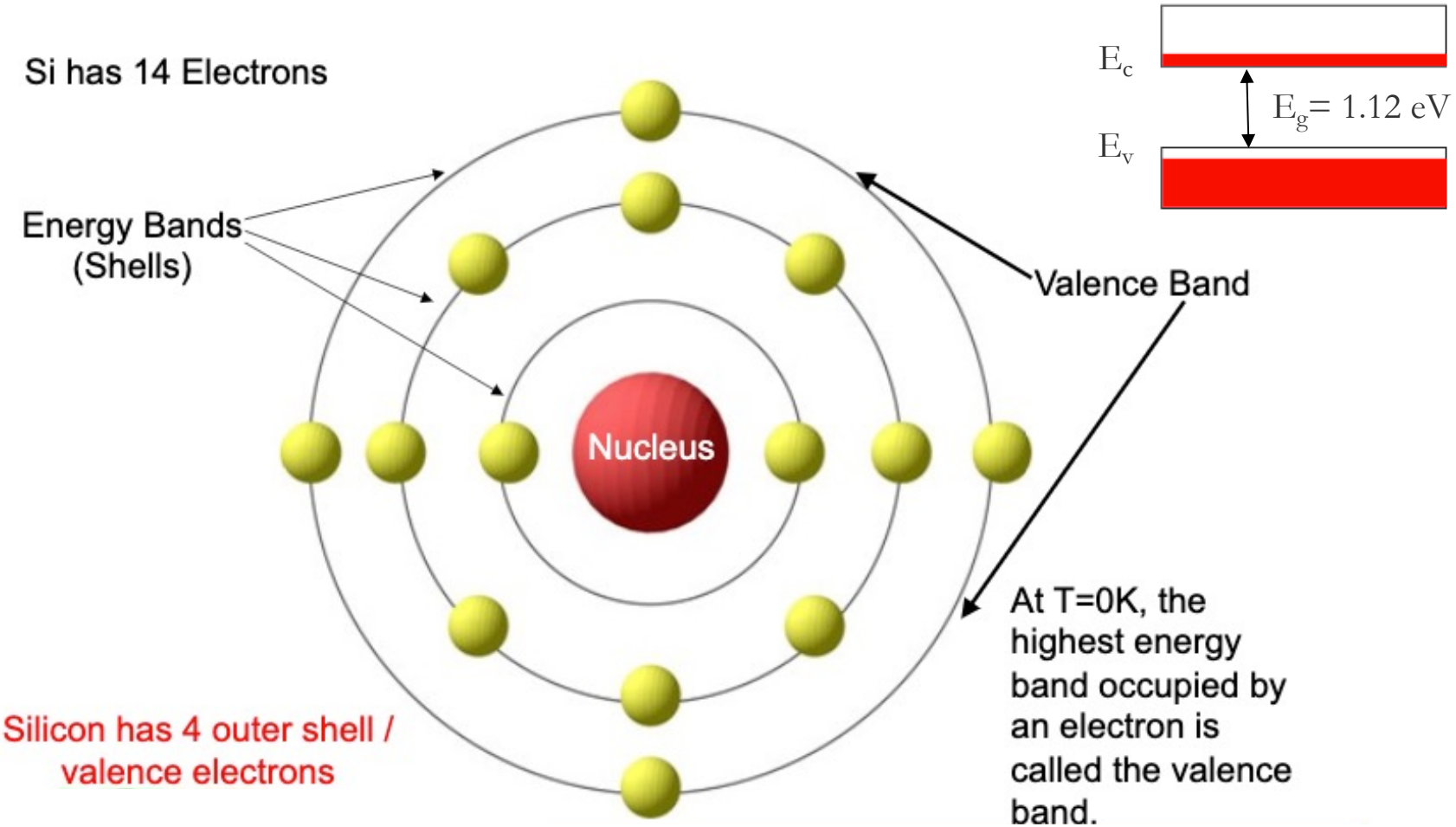
Semiconductor	Bandgap Energy E_G (eV)
Carbon (diamond)	5.47
Silicon	1.12
Germanium	0.66
Tin	0.082
Gallium arsenide	1.42
Gallium nitride	3.49
Indium phosphide	1.35
Boron nitride	7.50
Silicon carbide	3.26
Cadmium selenide	1.70

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

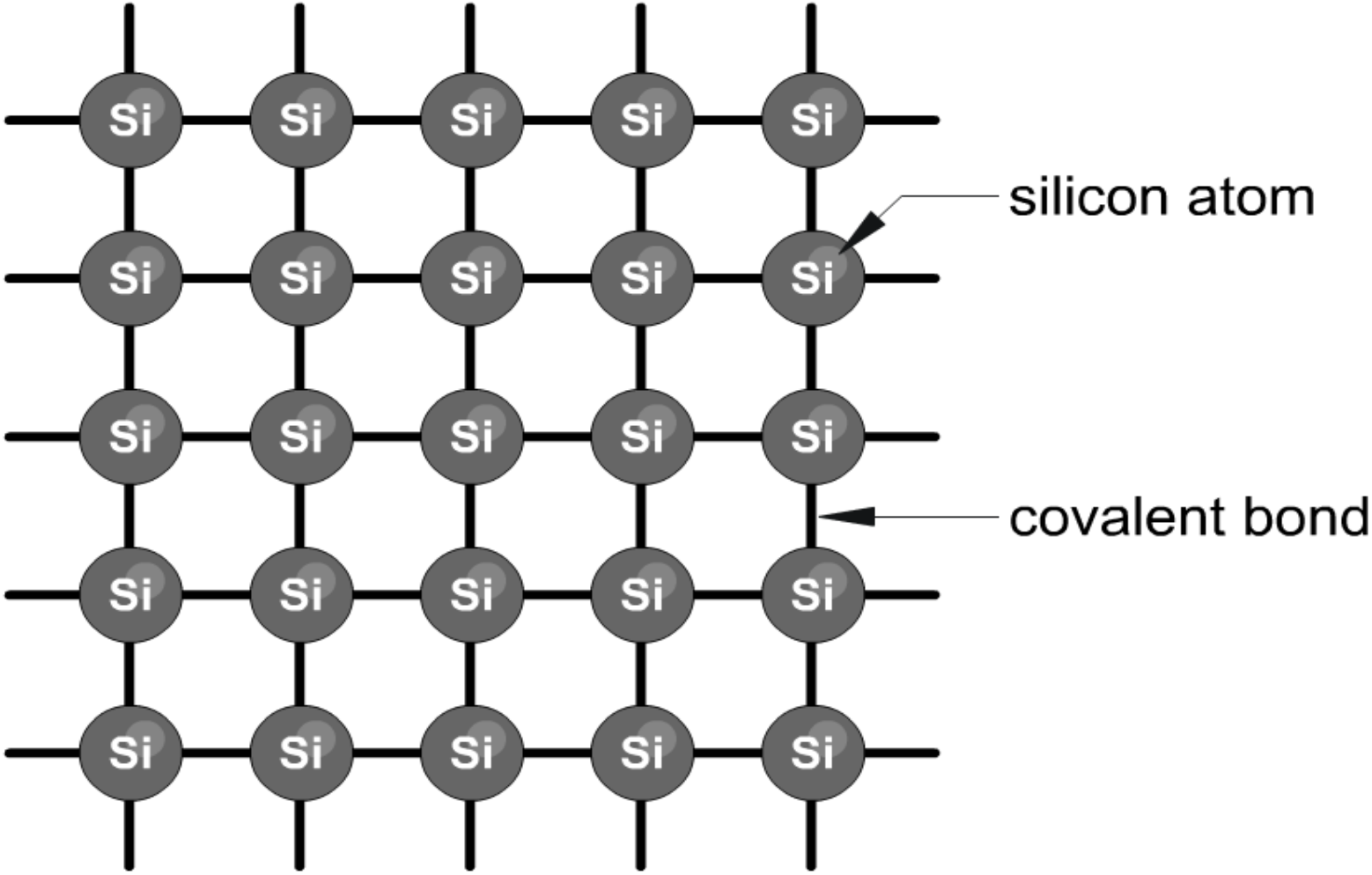
Example: Silicon



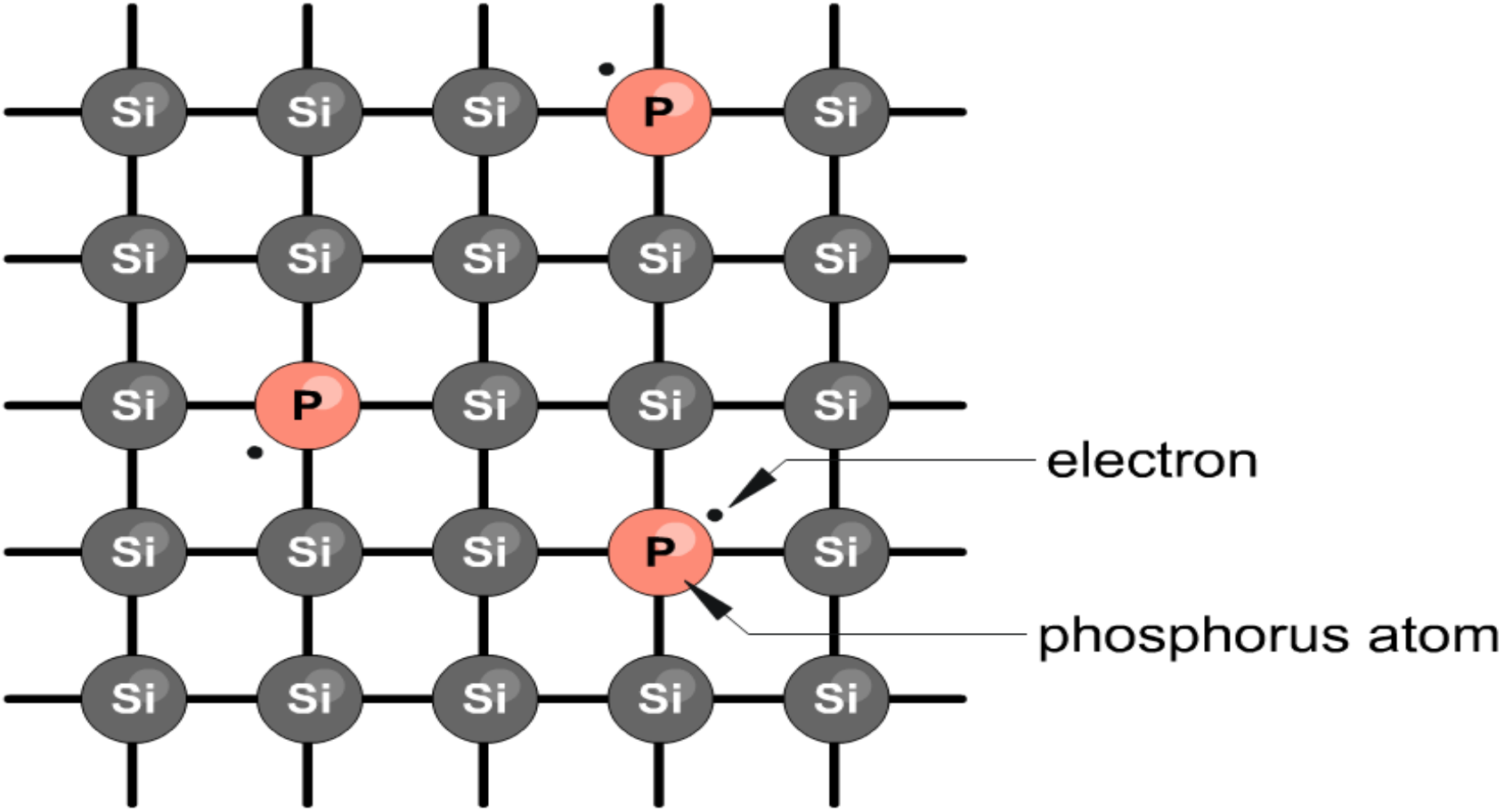
The Silicon Atom



Covalent Bonding in pure Si

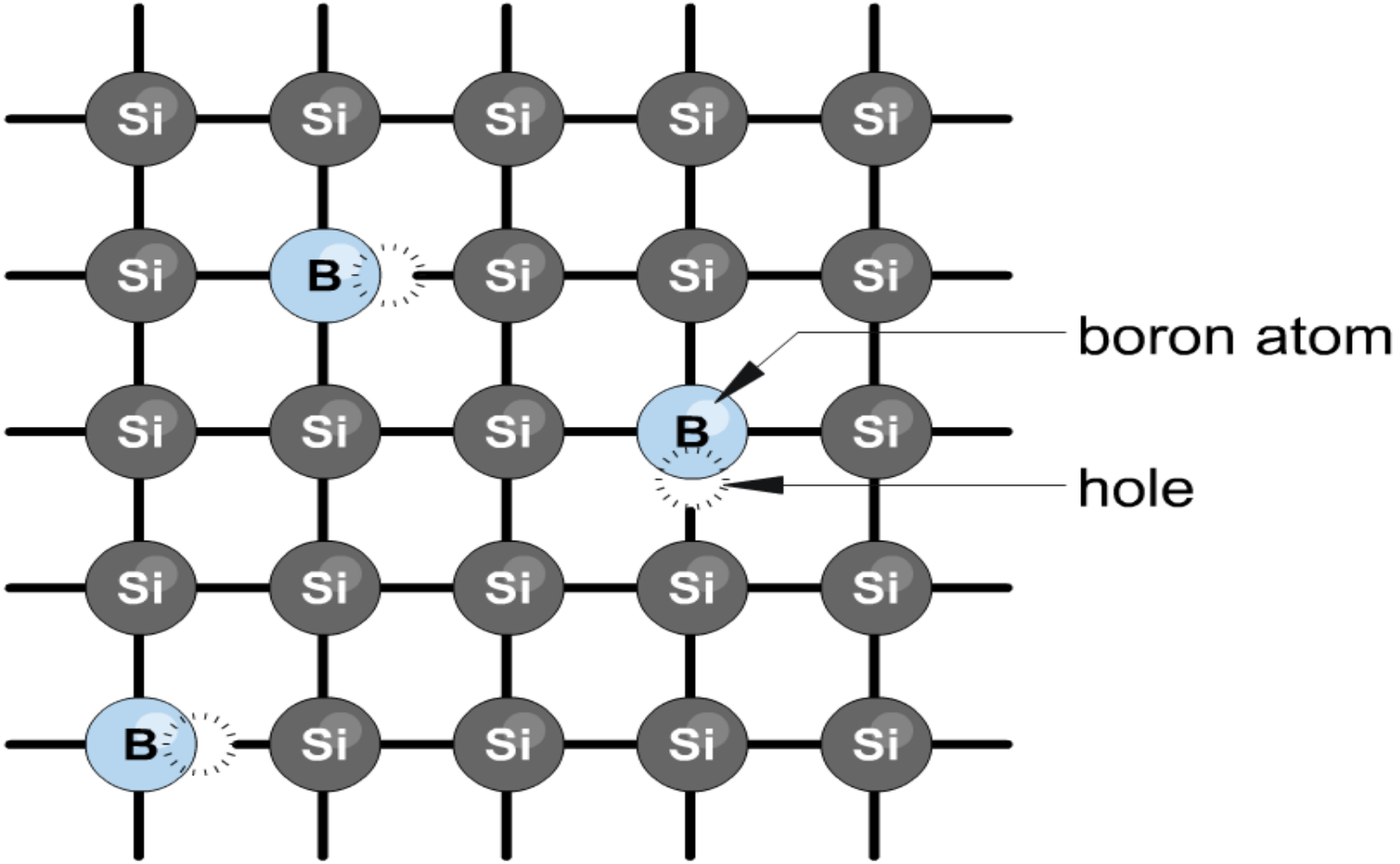


Doping: N-type Si



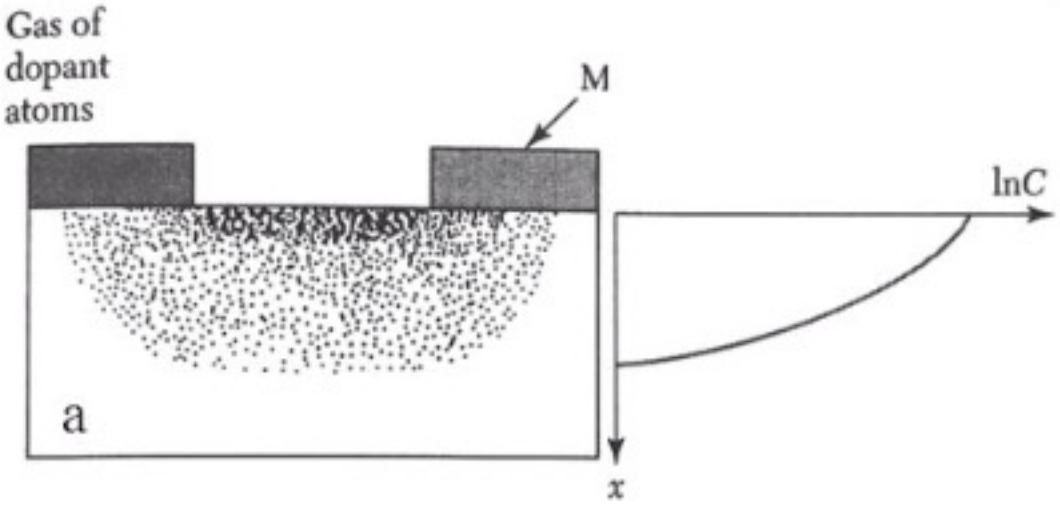
Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si



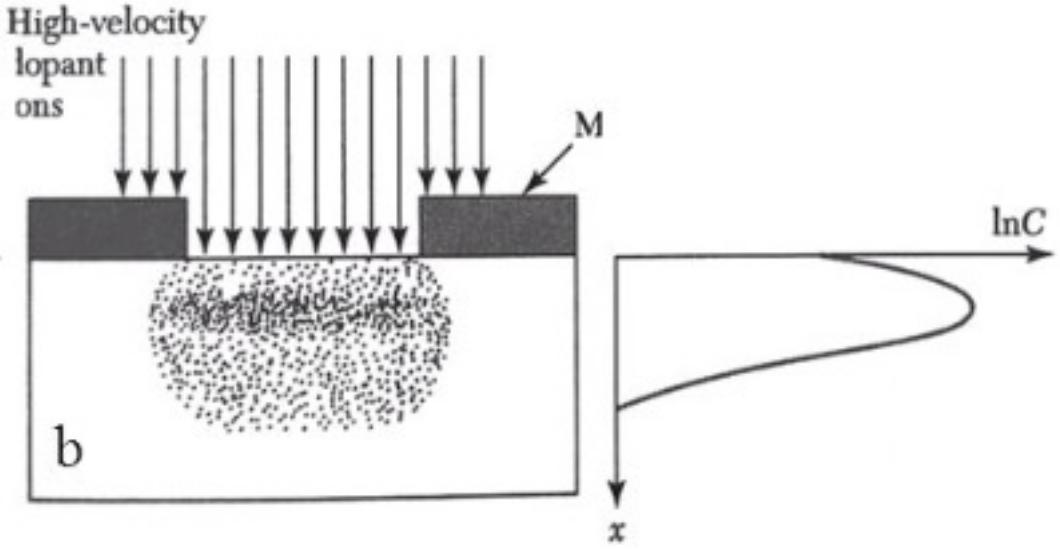
Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Methods of Doping: Diffusion



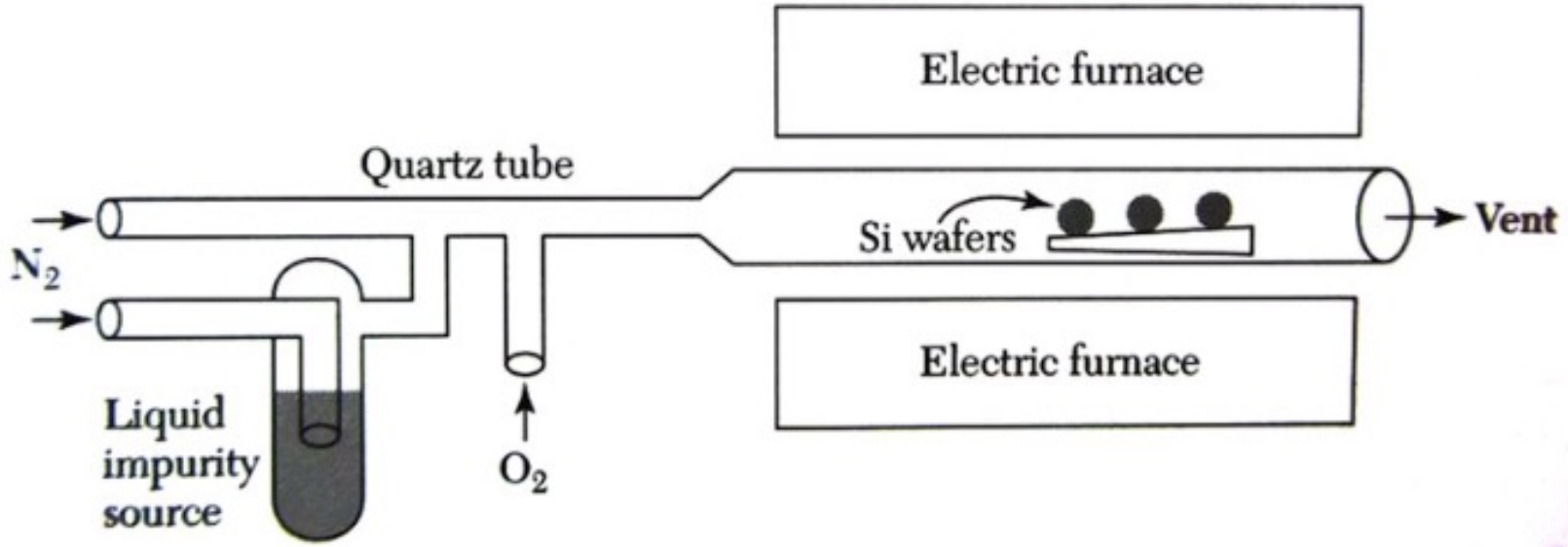
Diffusion is a cheaper and more simplistic method, but can only be performed from the surface of the wafers. Dopants also diffuse unevenly, and interact with each other altering the diffusion rate.

Methods of Doping: Ion implantation



Ion implantation is more expensive and complex. It does not require high temperatures and also allows for greater control of dopant concentration and profile. It is an anisotropic process and therefore does not spread the dopant implant as much as diffusion. This aids in the manufacture of self-aligned structures which greatly improve the performance of MOS transistors.

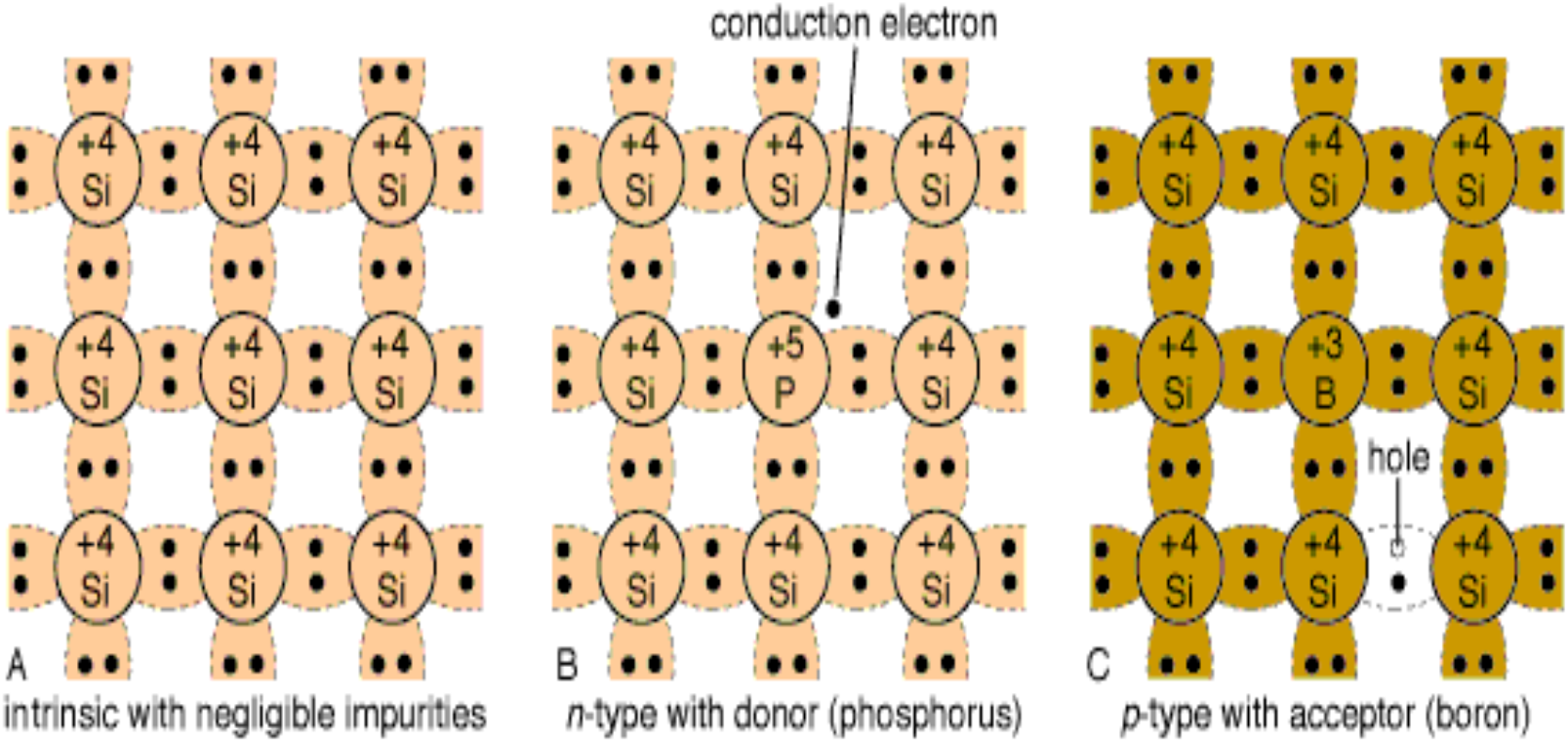
Example: N-type Si (Phosphorus doping)



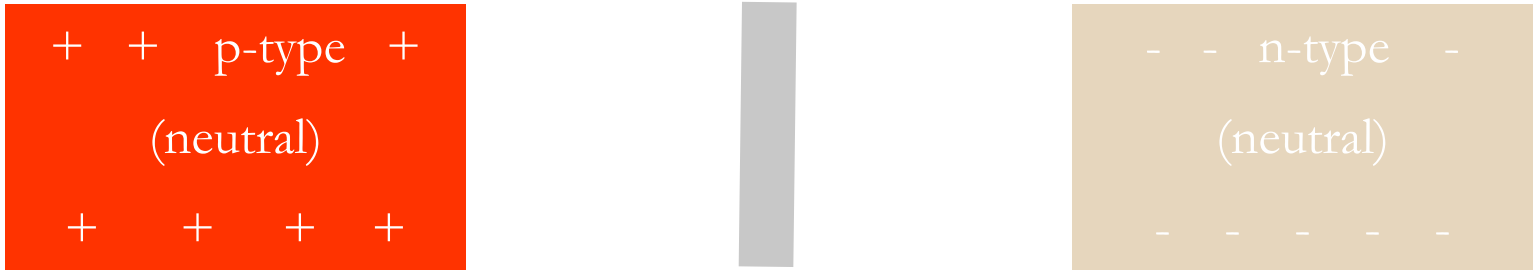
$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$

$$2P_2O_5 + 5Si \rightarrow 4P + 5SiO_2$$

Summary


conduction electron
+4 Si
+5 P
+3 B
hole
A
intrinsic with negligible impurities
B
n-type with donor (phosphorus)
C
p-type with acceptor (boron)

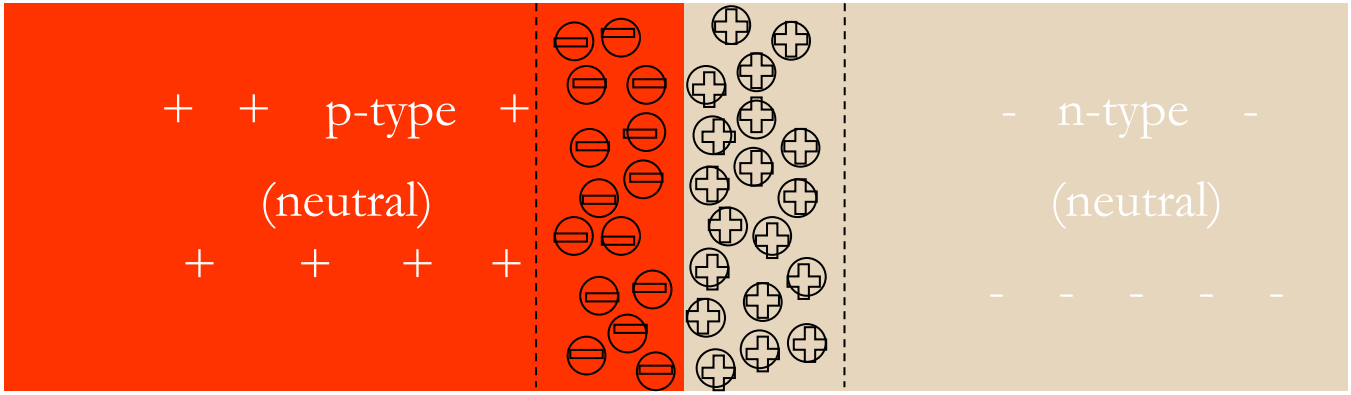
The PN junction



The basic structure of almost all active devices

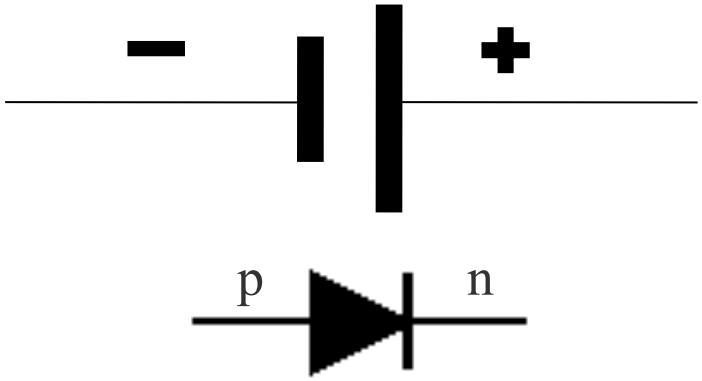
Diodes, transistors, lasers, photocells, solar cells, etc.

The PN junction

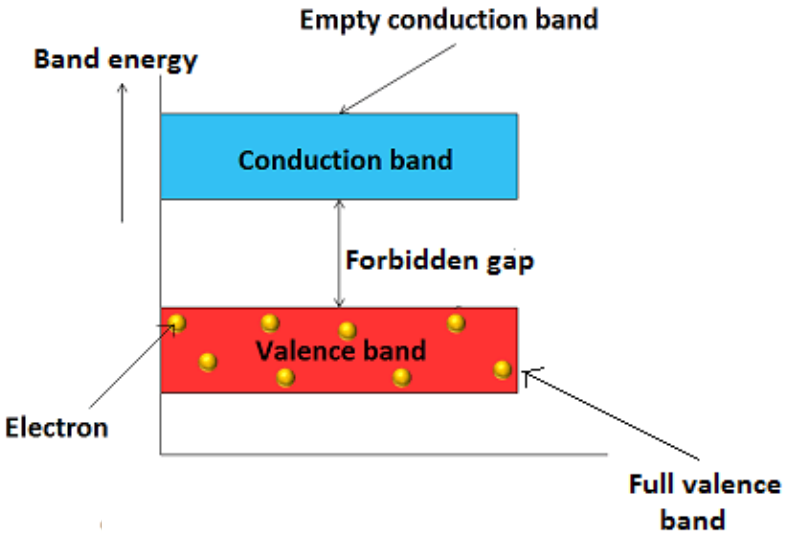


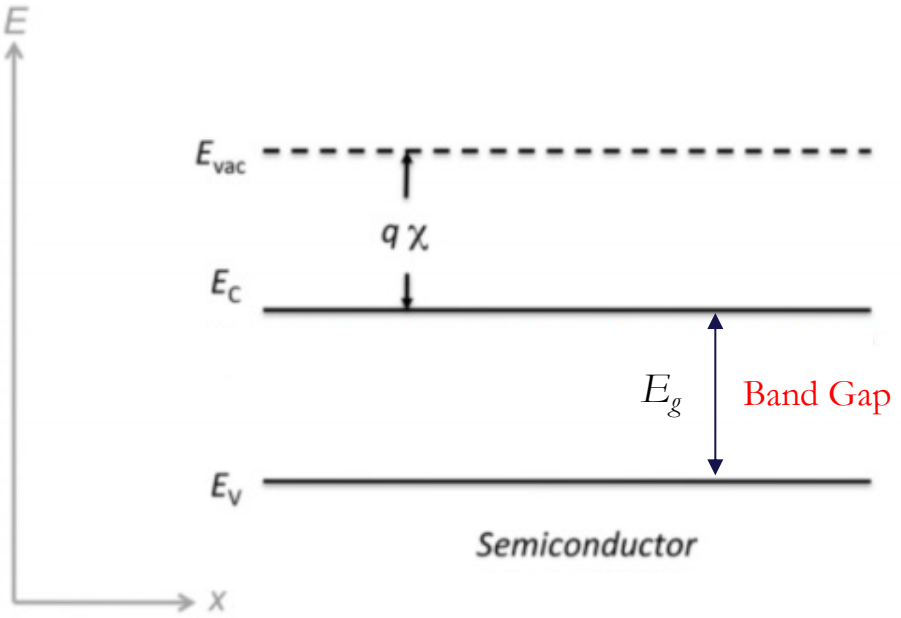
Depletion Region

Built-in Voltage (V_{bi})



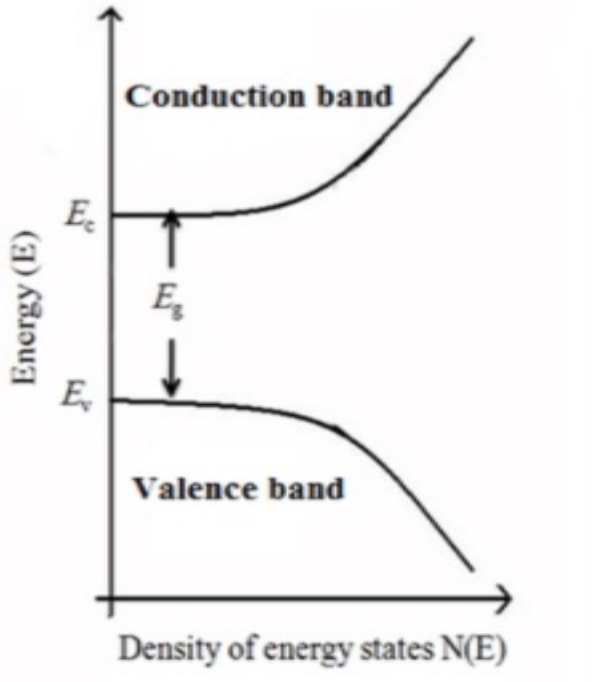
Back to the Energy Bands

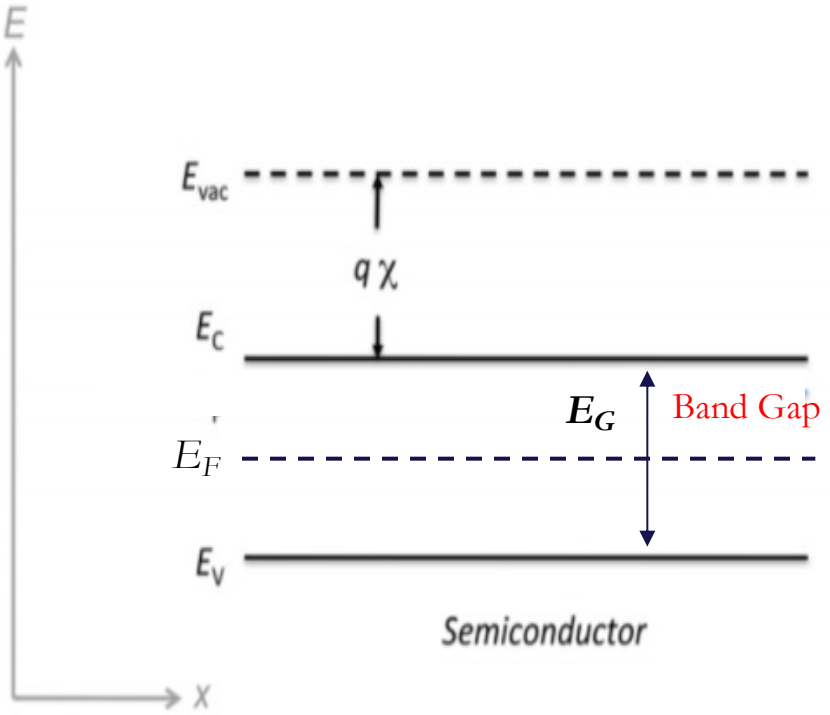




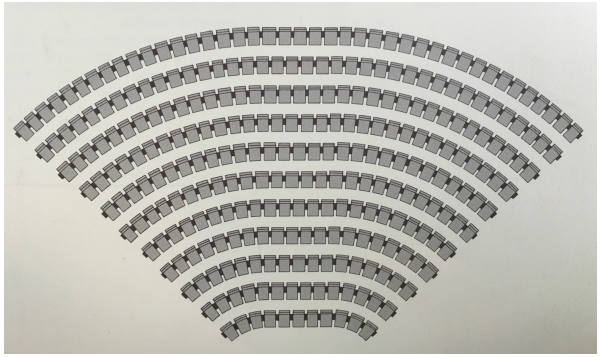
Band Diagram

Back to the Energy Bands



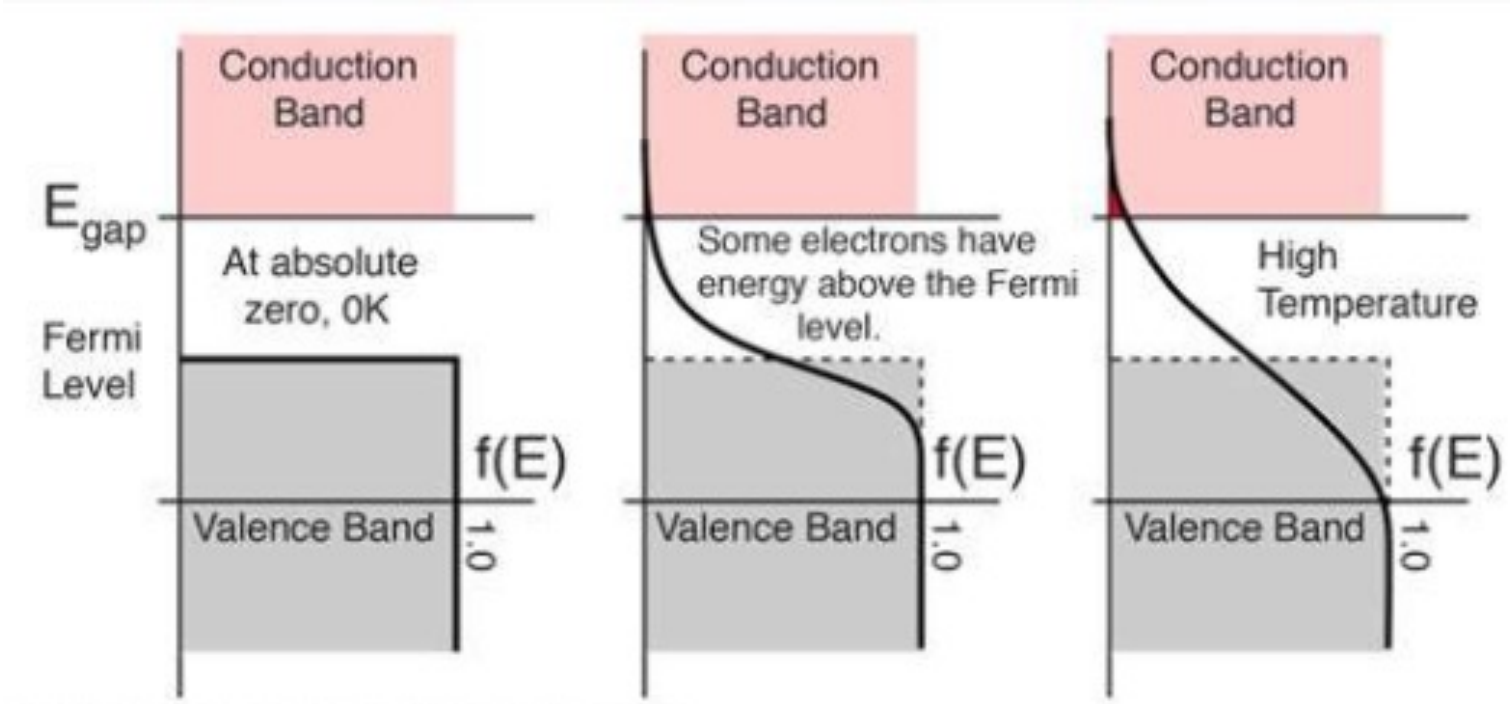


Density of States

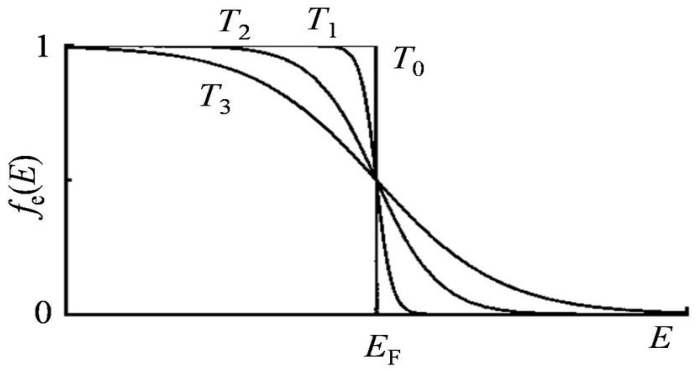


Band Diagram

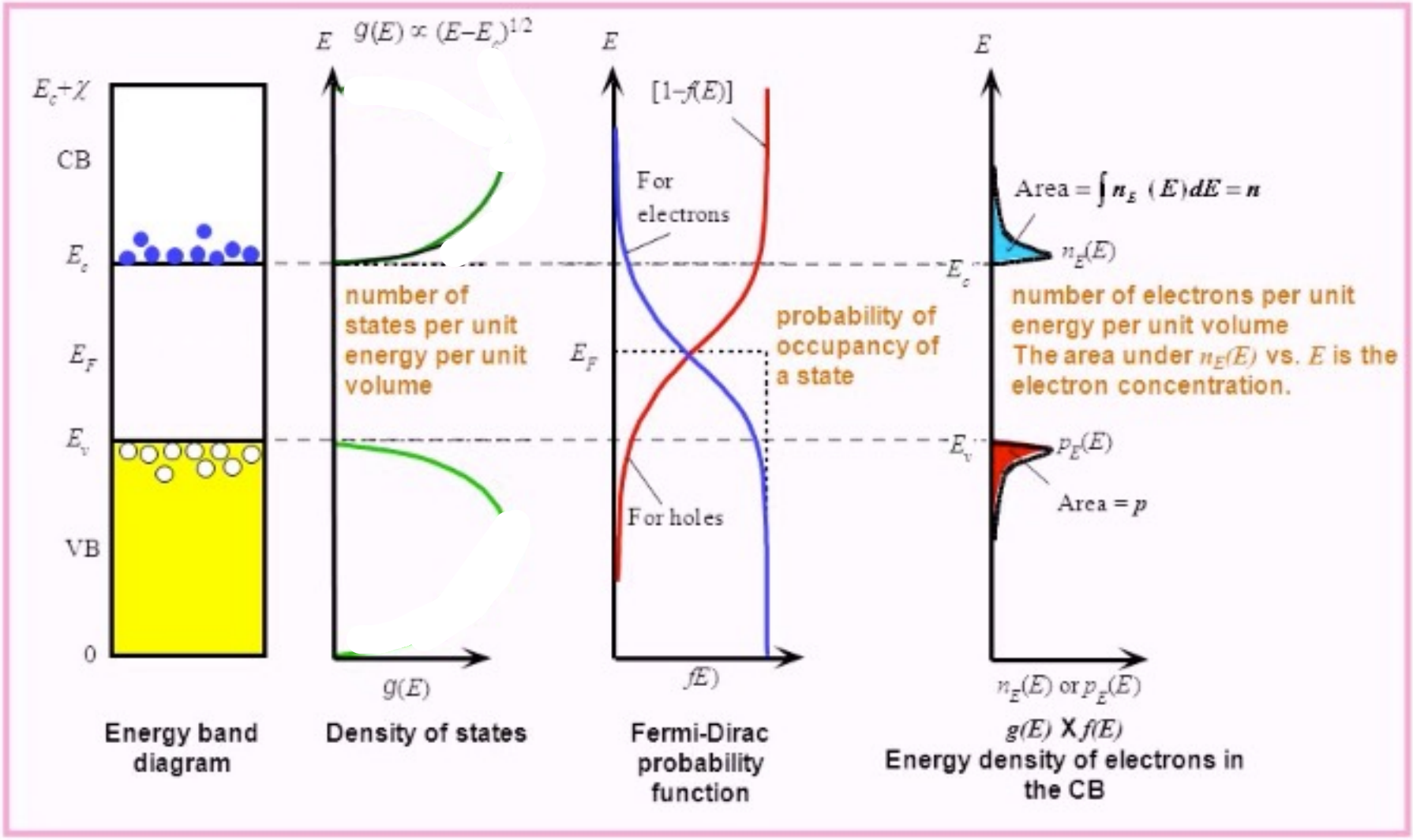
Fermi Distribution



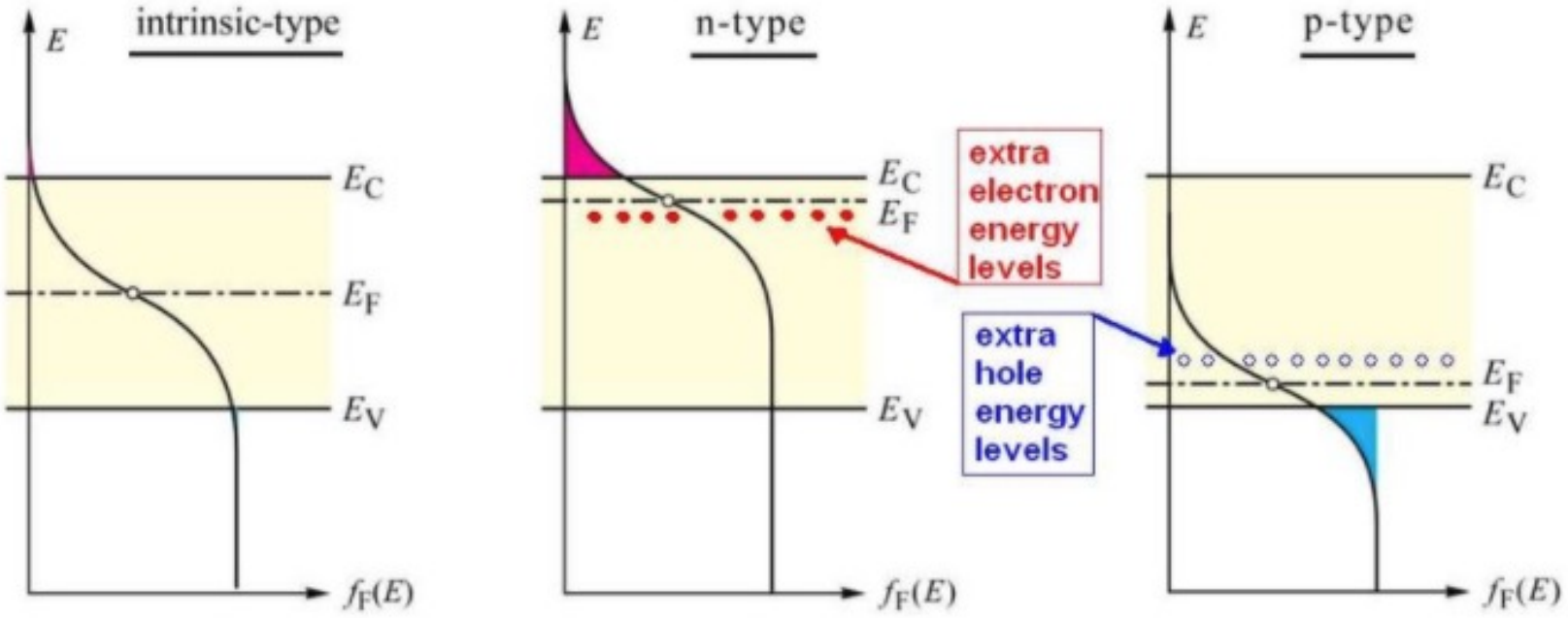
$$f_F(E) = \frac{1}{1 + exp\left(\frac{E - E_F}{kT}\right)}$$



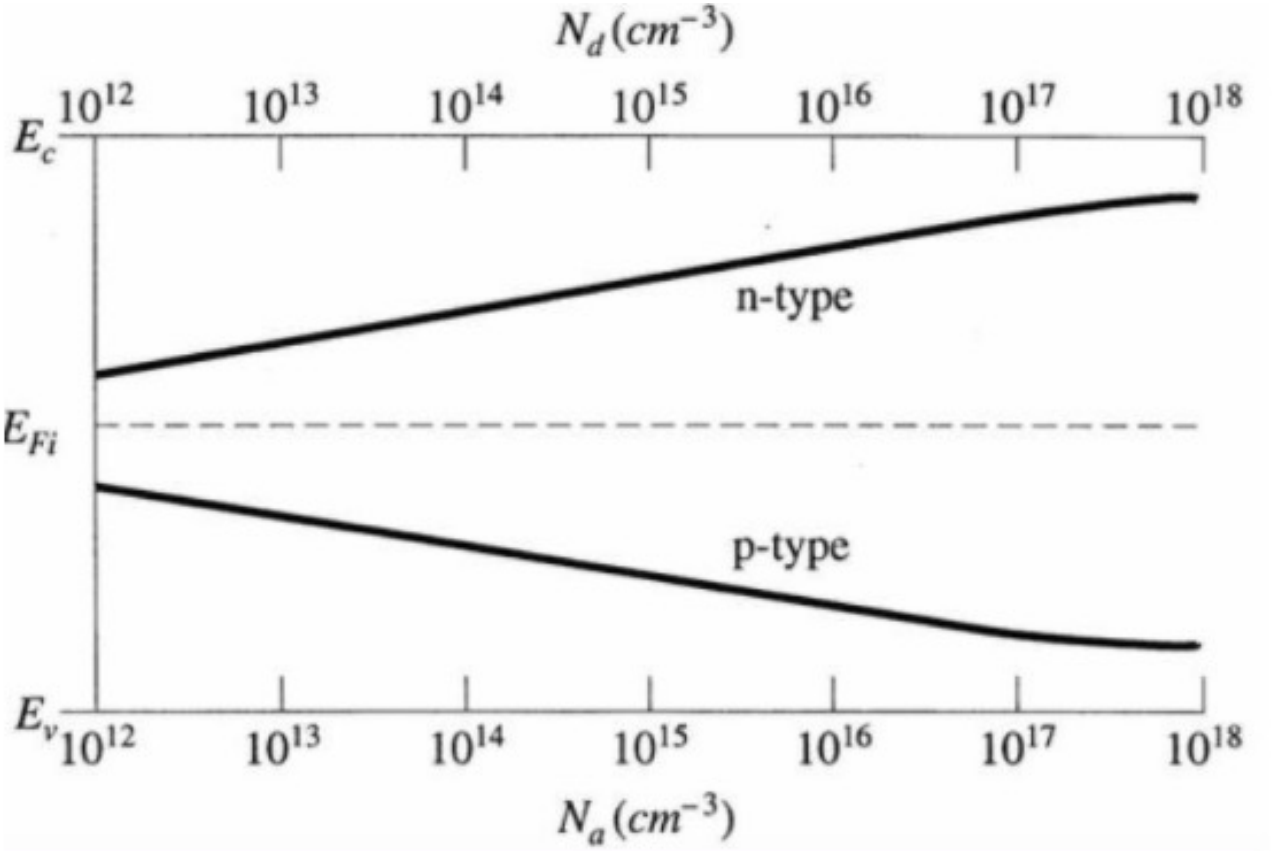
Electron/Hole Density



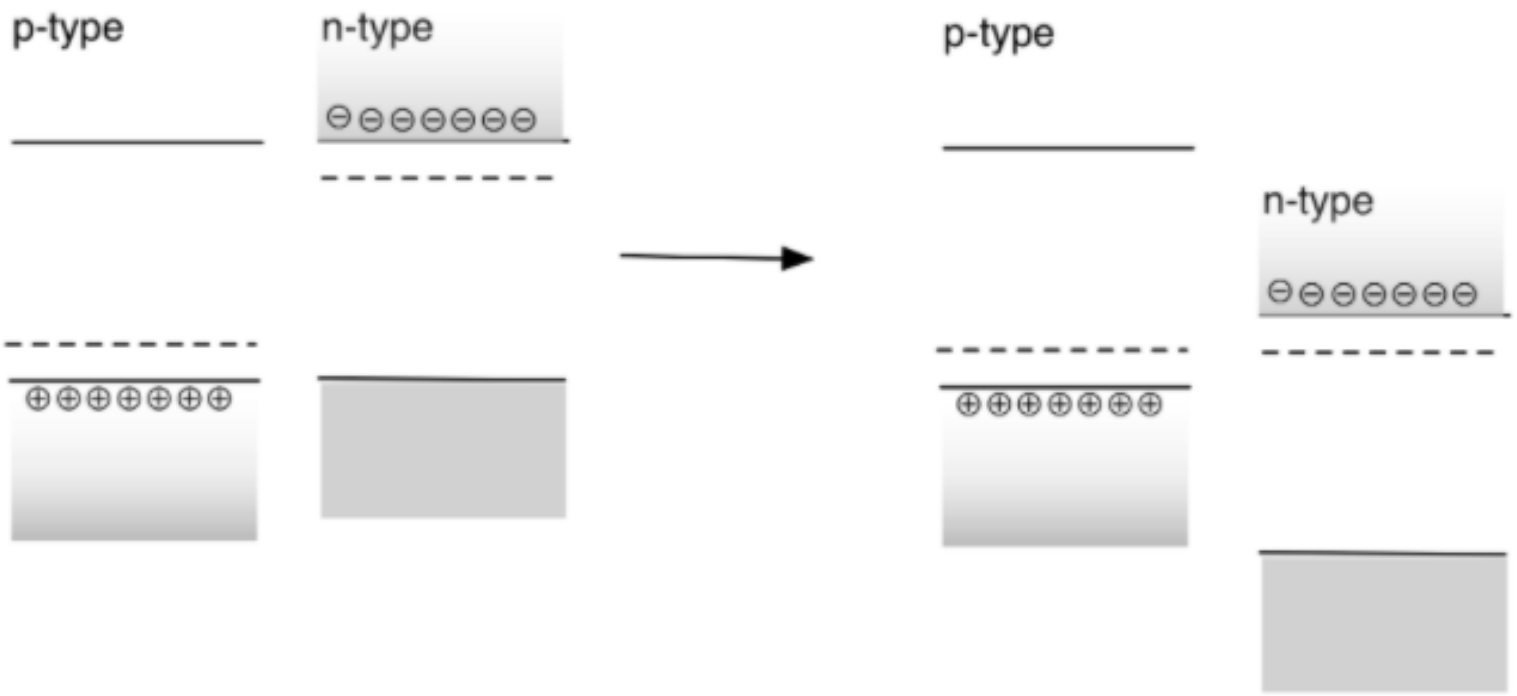
The n-type and p-type Band Diagram



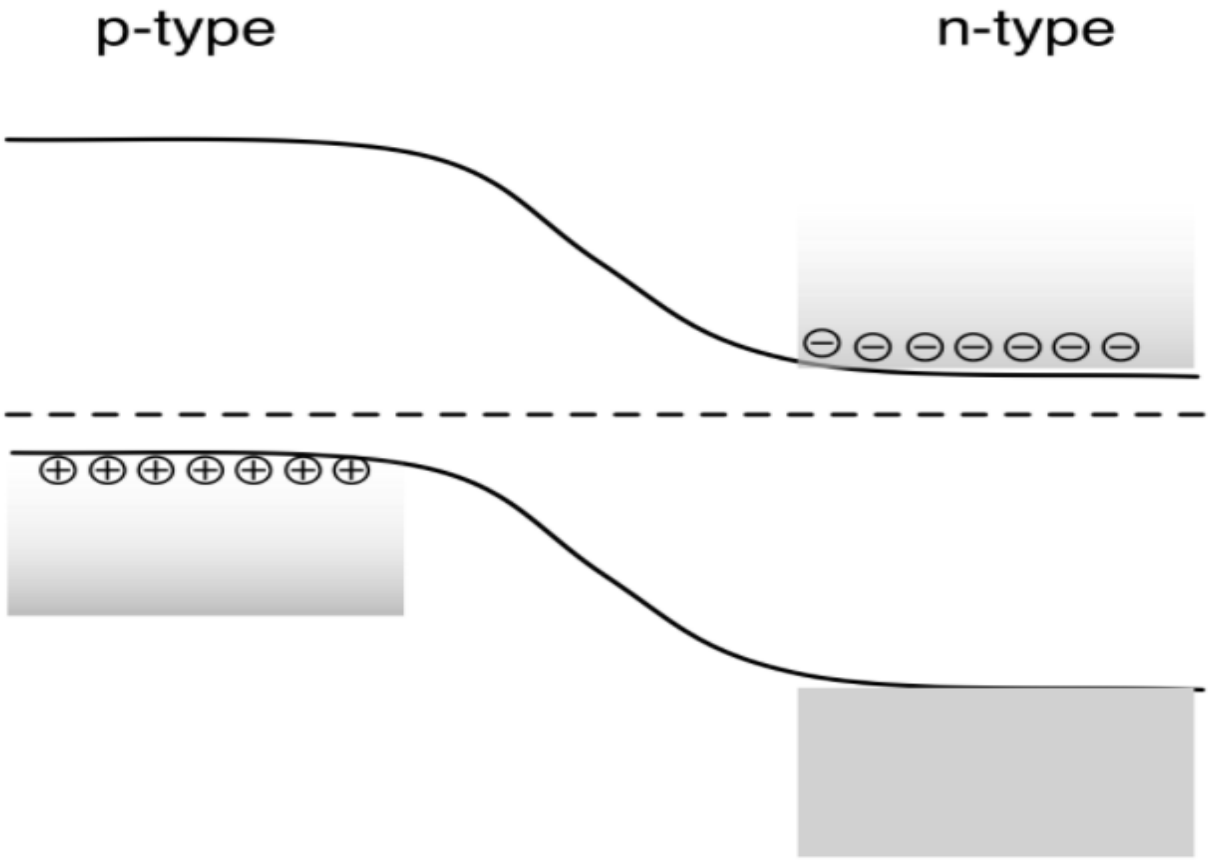
The n-type and p-type Band Diagram



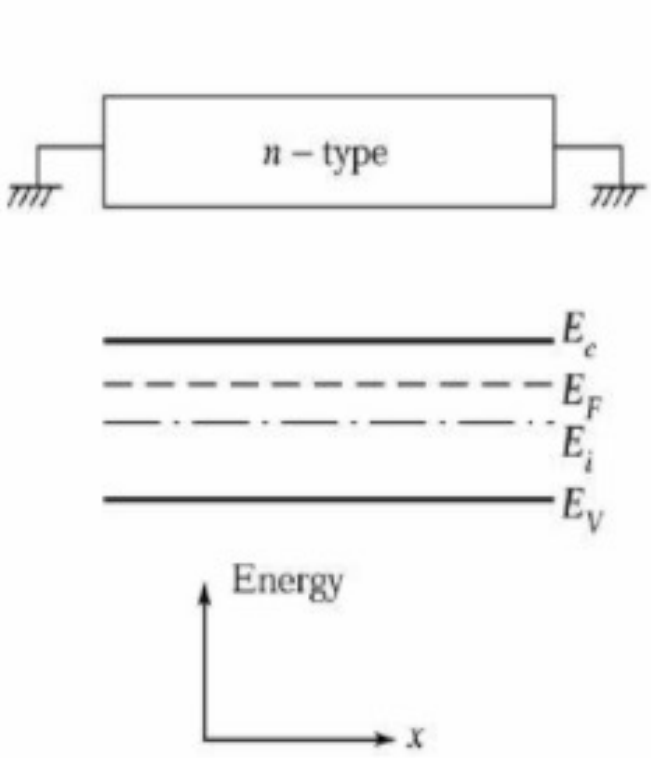
The PN Junction Band Diagram

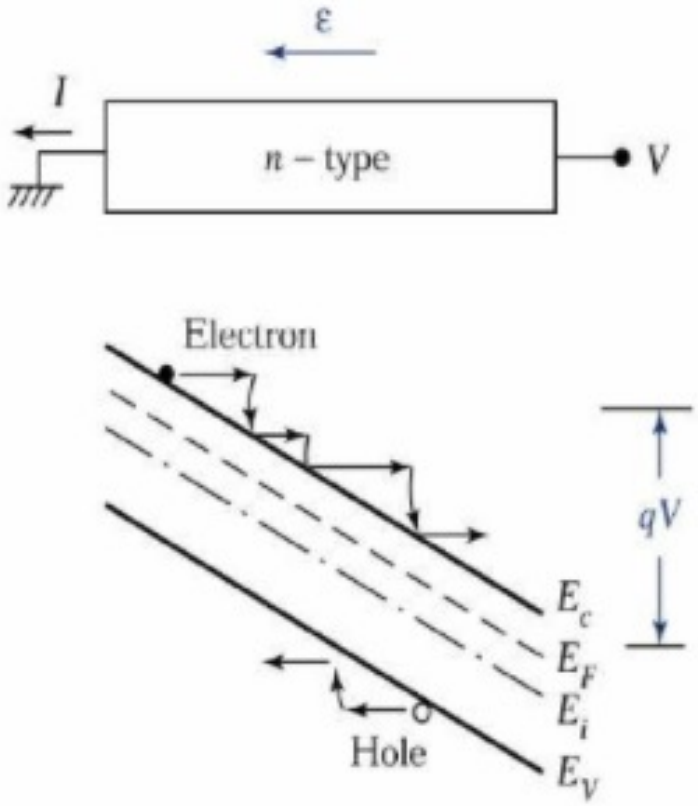


Unbiased PN Junction Band Diagram

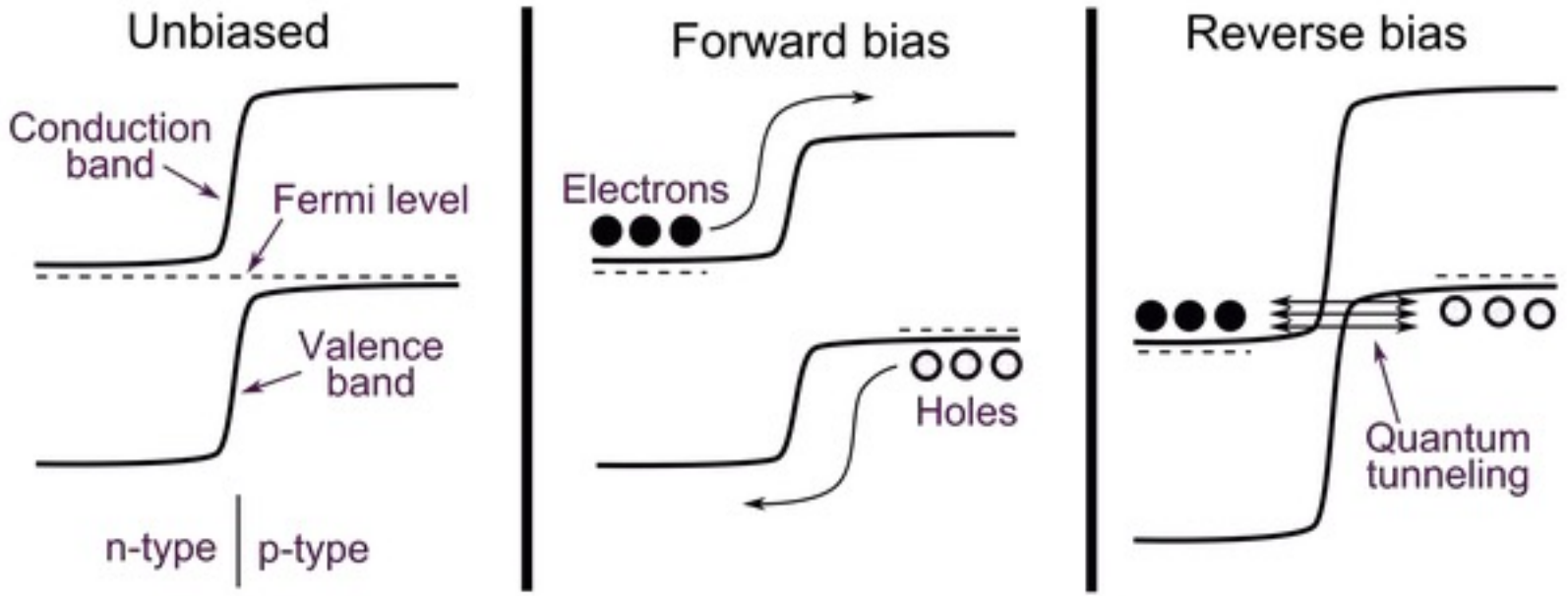


Biased n-type Band Diagram



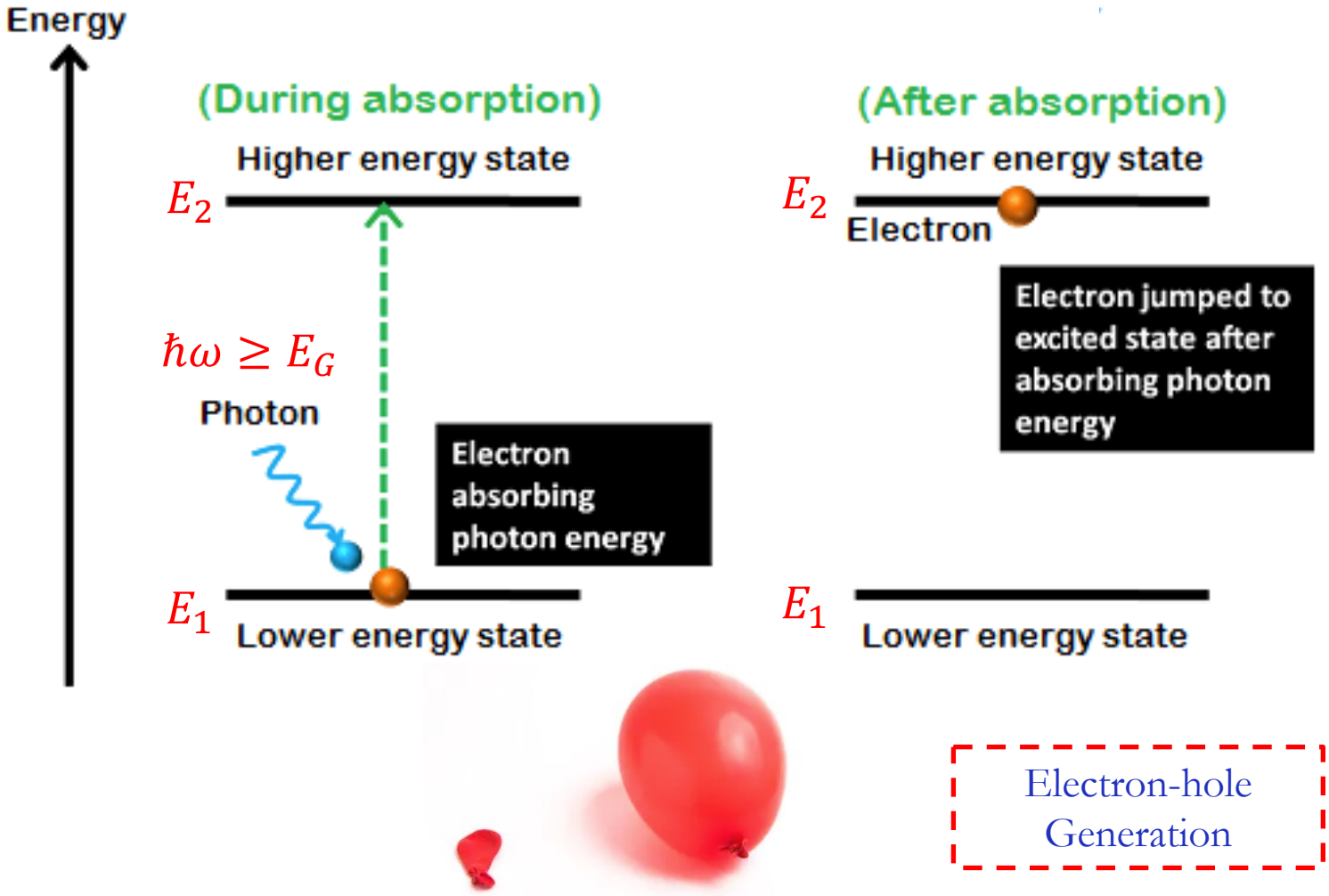


Biased PN Junction Band Diagram



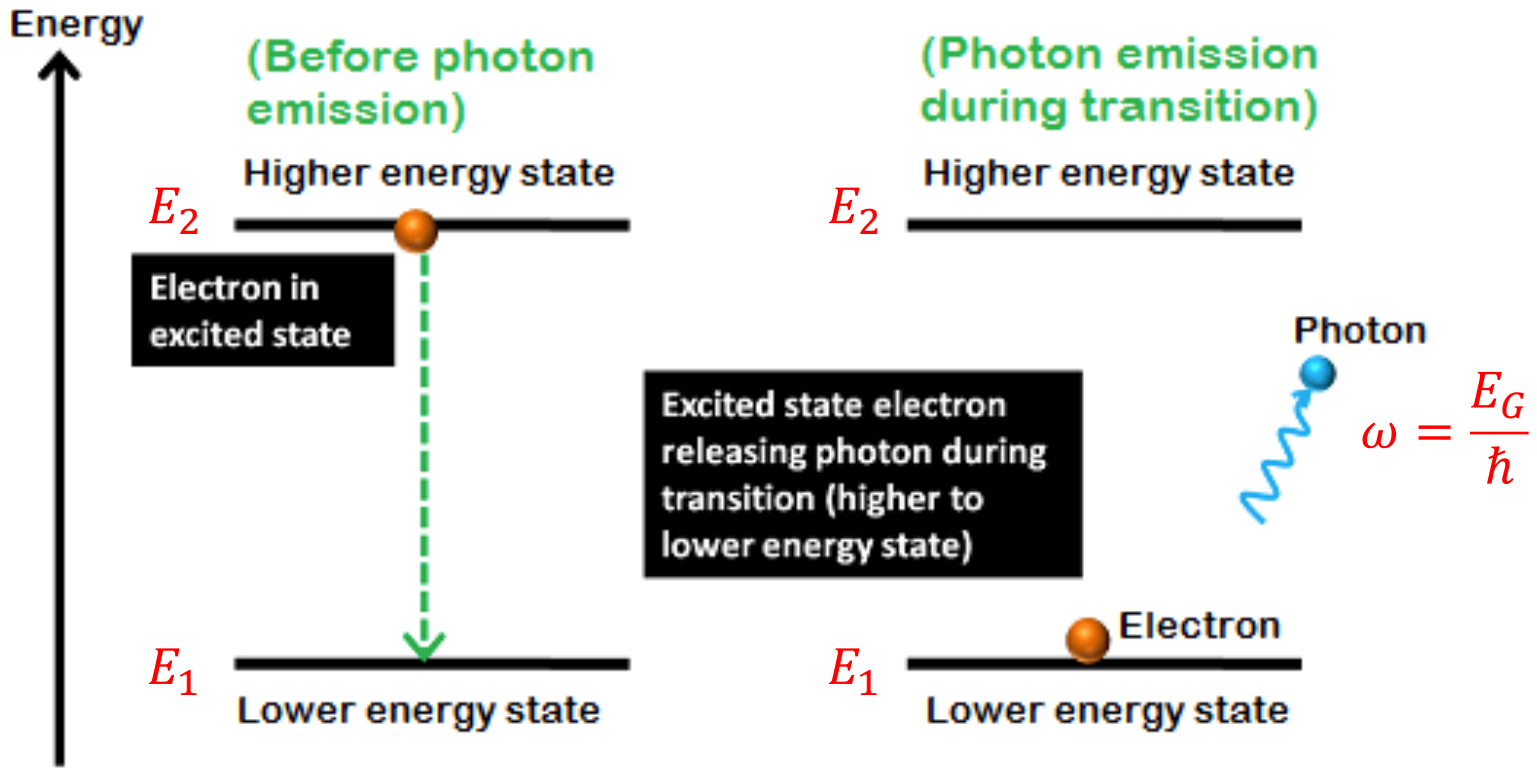
Photon Generation and Absorption

Absorption



Photon Generation and Absorption

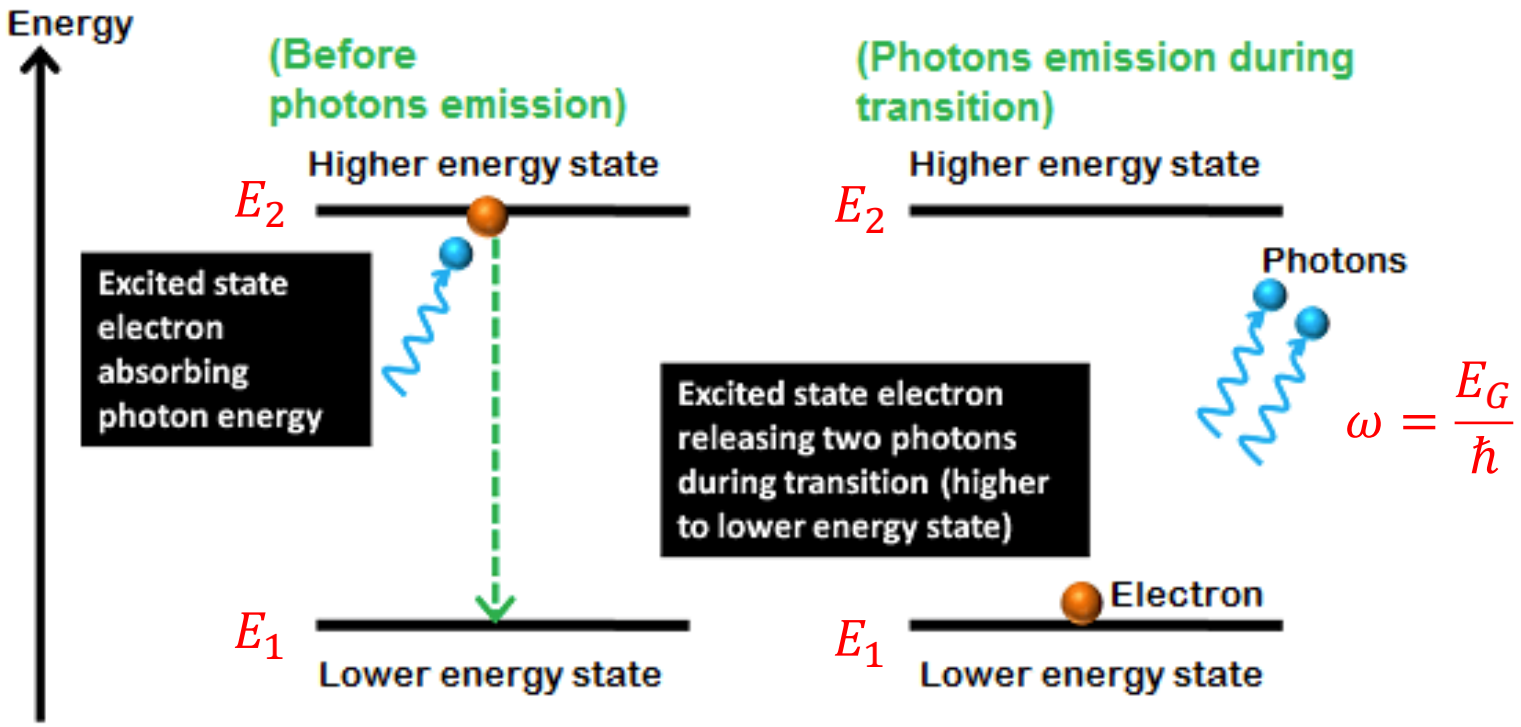
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

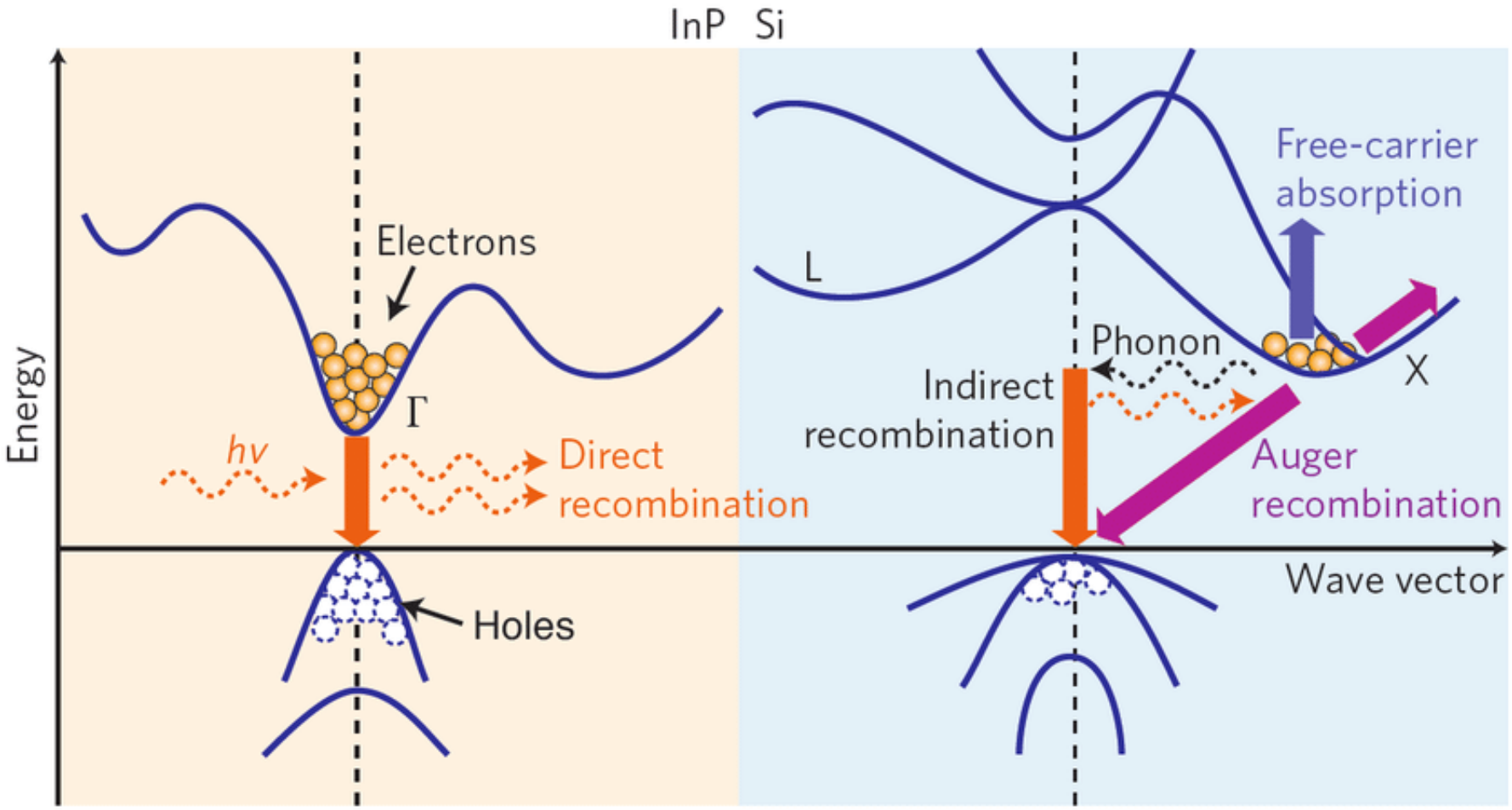
Photon Generation and Absorption

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

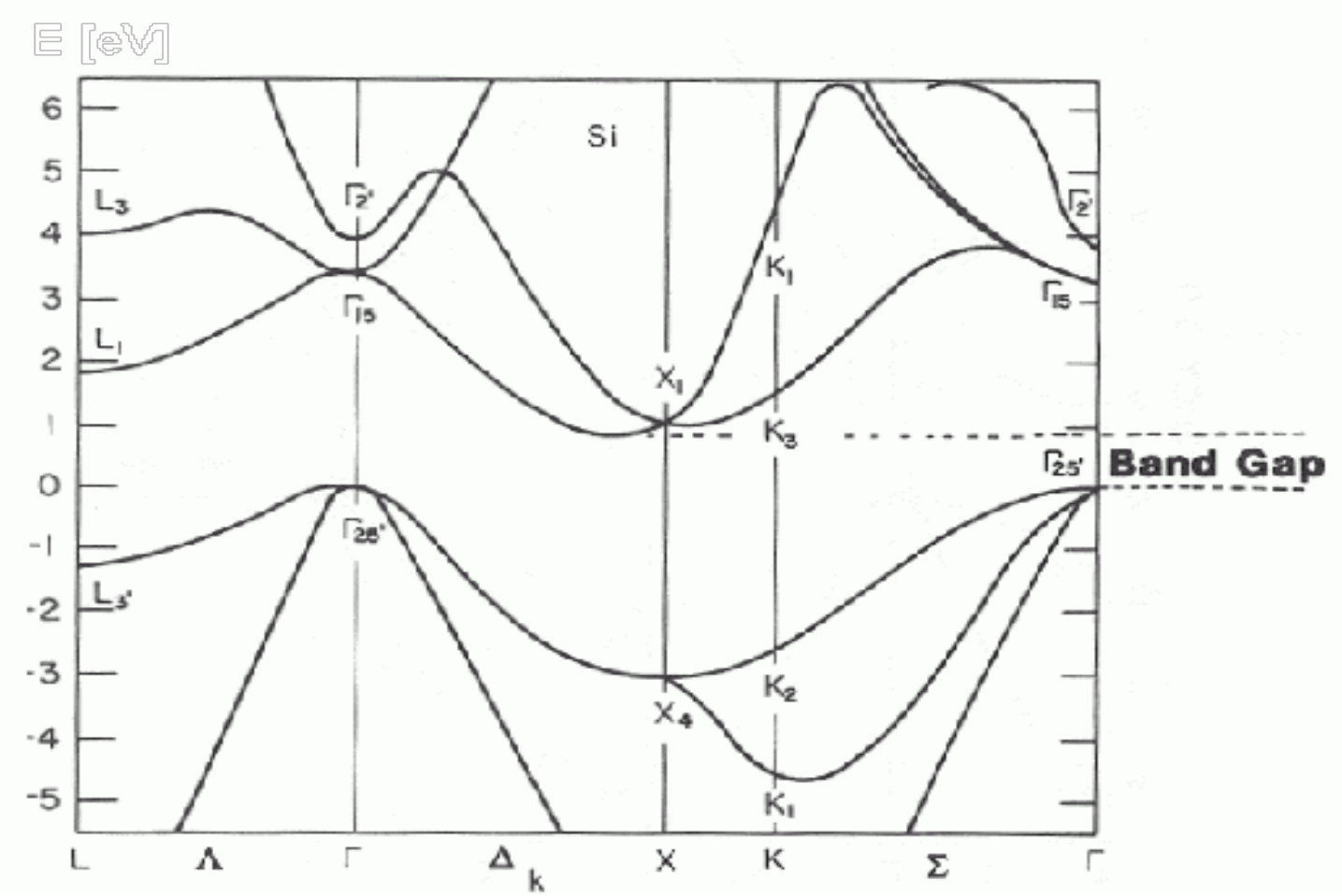
Direct and Indirect Gap Materials



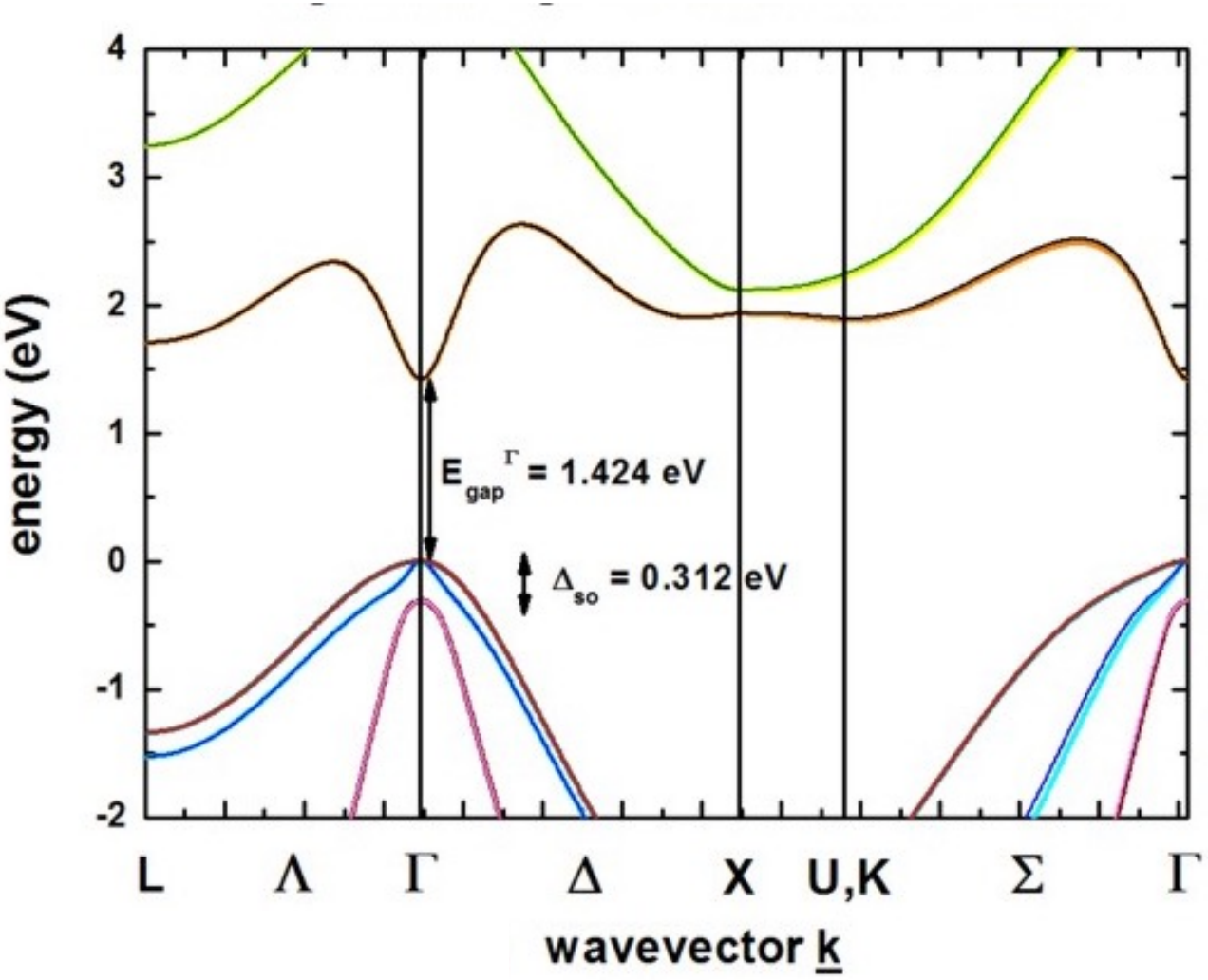
Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

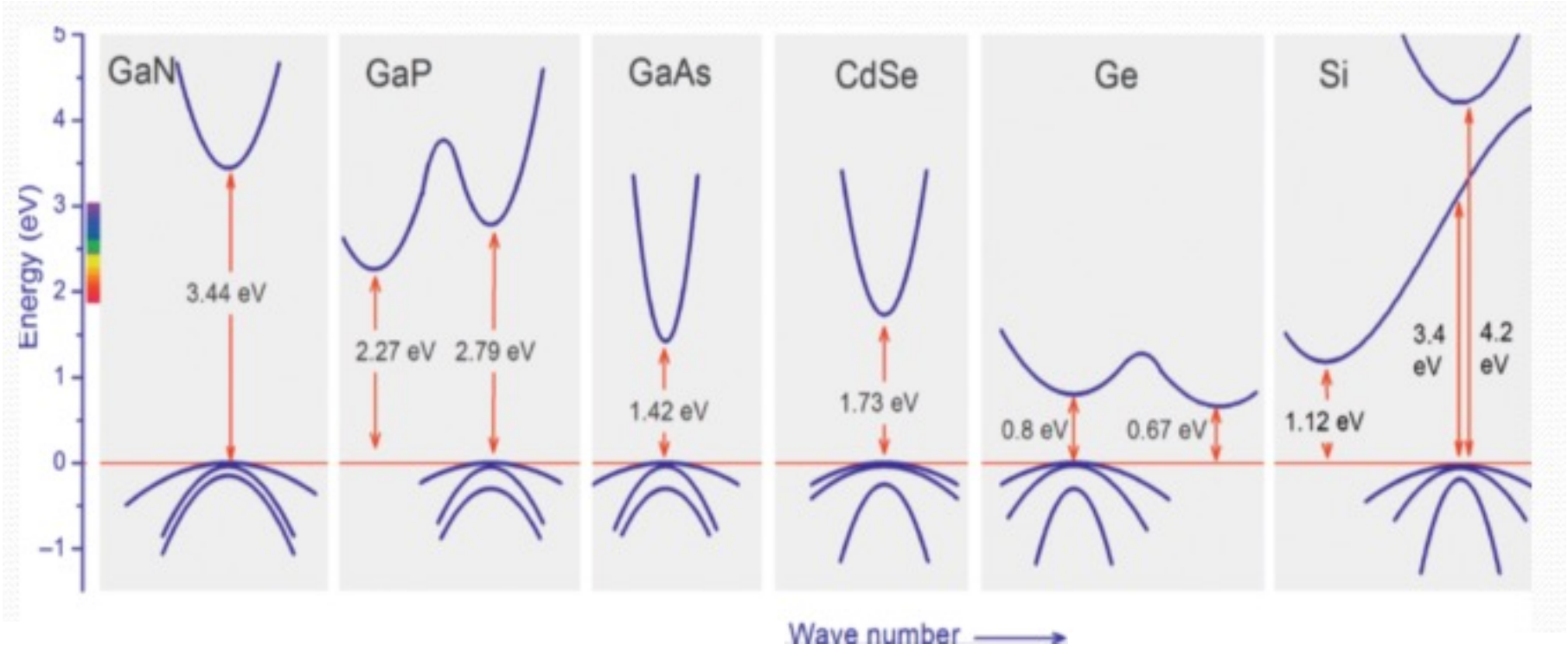
Si: Indirect Gap Material



GaAs: Direct Gap Material



GaAs: Direct Gap Material



Bandgap Engineering

Example: Emission in GaAs, GaN and InN.

$$\omega = \frac{E_G}{\hbar}$$

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi c\hbar}{E_G}$$

$$\lambda_g = \frac{hc}{E_G}$$

GaAs band gap energy E_G=1.42 eV

$$\lambda_g = \frac{2\pi(3\times10^8)(1.055\times10^{-34})}{(1.42)(1.6\times10^{-19})}$$

$\lambda_g = 875\ nm$ NIR

GaN band gap energy E_G=3.49 eV

$\lambda_g = 356\ nm$ UV

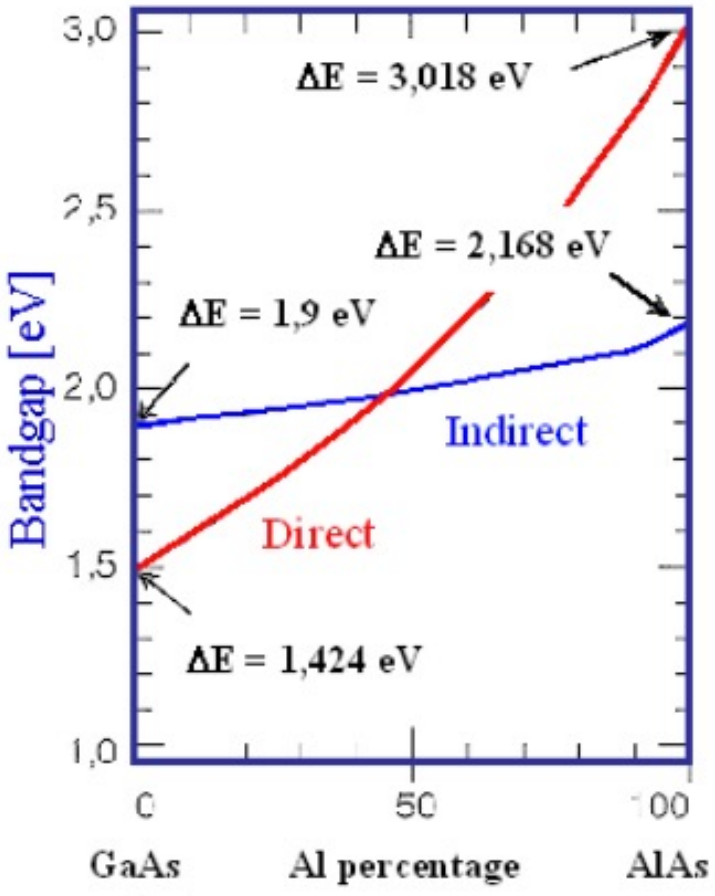
InN band gap energy E_G=0.69 eV

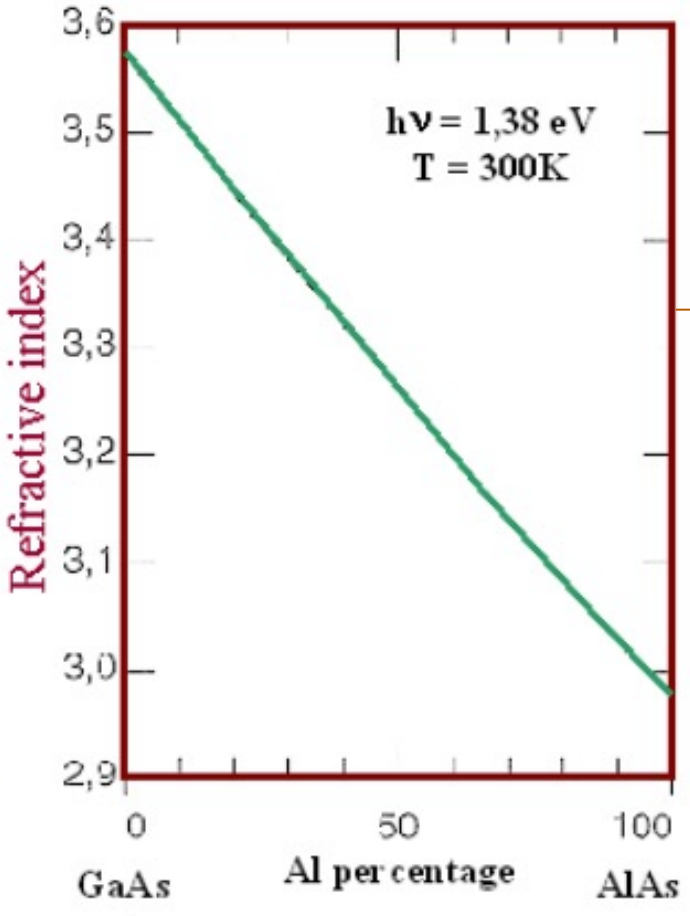
$\lambda_g = 1800\ nm$ IR

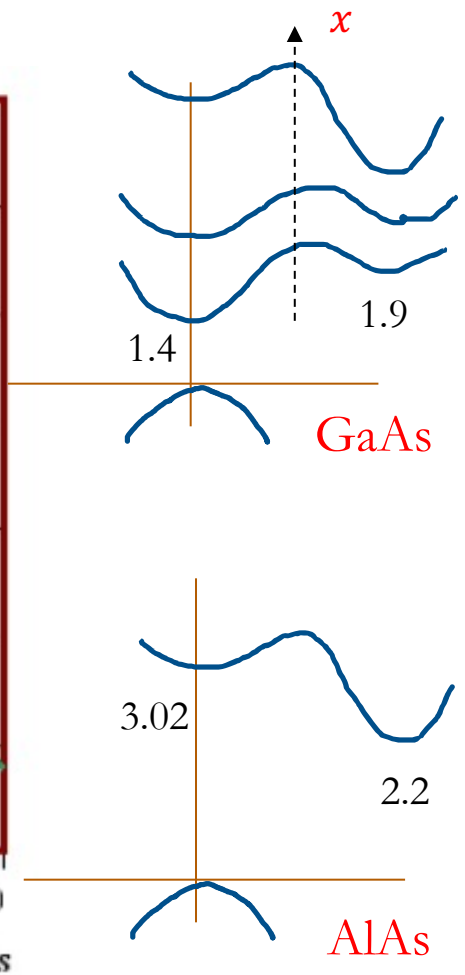
The alloy $In_xGa_{1-x}N$ is very useful for spanning the visible range.

Bandgap Engineering

Example: Emission in $Al_xGa_{1-x}As$.





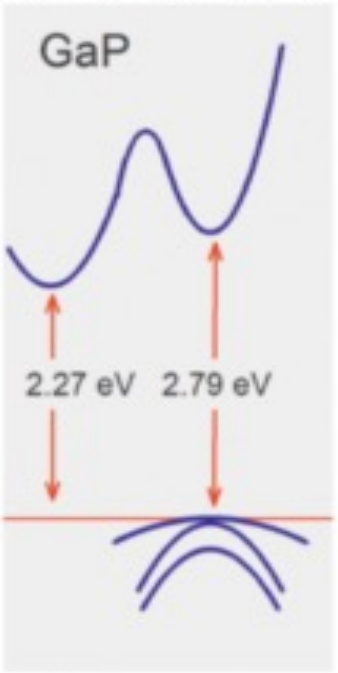


The Green Gap

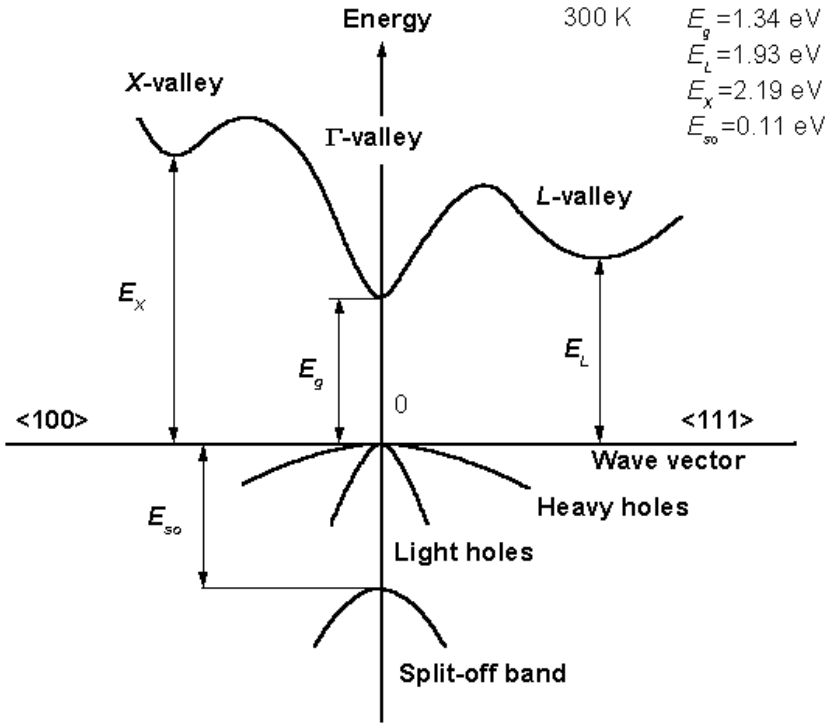
Green Band: 530-570 nm.

Green Band: 2.2-2.3 eV.

Emission in $Ga_xIn_{1-x}P$.



InP



Q & A

Send me an email if you have any questions.